

Boundedly Rational Information Demand

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Abstract

Information is valuable because it can change future choices, so its value should increase with both the probability of changing the choice and the payoff gain when it does. I test these comparative statics in experiments where participants choose between two lotteries and evaluate information that reveals one lottery's outcome. Elicited valuations deviate sharply from the rational benchmark: they often show insensitive or wrong-signed comparative statics, and seemingly correct patterns sometimes reflect context-dependent heuristics. Diagnostic treatments show that salience, contingent-choice planning, and simpler computation help only modestly; large deviations remain unless valuation no longer requires integrating payoffs from multiple choices.

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1 Introduction

In economics, information is valued primarily for its ability to guide future decisions. For example, investors read analyst reports to determine which stocks to buy, consumers read product descriptions and online reviews to find the most suitable product, and recruiters conduct interviews to identify the best job candidates. Despite the fact that information value should heavily depend on the decision problem it informs, we know surprisingly little about their empirical relationship.

This paper studies the demand for information about choice options in an experimental setting, with particular focus on how this demand responds as the prior beliefs about these options change. This question is important for several reasons. First, many models in information economics assume that information demand correctly responds to incentives. Testing this assumption informs the realism of theoretical results. Second, information acquisition directly affects decision quality and overall payoffs. Hence, analyzing information demand across different decision problems helps us identify decisions most likely to be suboptimal. Third, in market environments, buyers' information acquisition affects the purchase probability of each product. As a result, how information demand responds to the prior beliefs of the products affects sellers' incentive to invest in them. Lastly, in settings such as recruitment, admission, and lending, information acquisition based on prior beliefs about applicants carries implications for diversity, equity, and inclusion.

Demand for information is determined by its valuation, which has a well-defined rational benchmark. According to standard information economics, information has value only if it steers choice from the default option to an alternative with some probability. Moreover, the exact value of information is determined by the probability of choosing an alternative and its value-added when chosen. Formally, consider an expected utility maximizer who needs to choose from a set of options. Each option's utility u depends on the unobserved state of the world ω . The value of an information structure I (a distribution over a set of signals S_I correlated with the states) can be calculated by:

$$V(I) = \sum_{s \in S_I} \mathbb{E}[u(a(s), \omega) - u(d, \omega) | s] \cdot p(s) \quad (1)$$

where d is the default option chosen without the information, $a(s)$ is the alternative chosen if signal s is realized, and $p(s)$ is the probability of s . This expression implies two key comparative statics. First, for each signal that induces an alternative different from the default, the higher the value-added of the induced choice, the greater the information value. Second, the higher the probability of such a signal, the greater the information value.

In this paper, I study the demand for information about choice options in a simple experiment using these two comparative statics as the rational benchmark. Participants face a choice between two independent binary lotteries, D and A. Each lottery pays out \$3 if it wins and \$0 otherwise. Their winning chances are known and denoted by d and a , respectively. Lottery D is more likely to win, so it is the default option that participants should choose without additional information. In the main treatments, D-Info, participants may receive information that fully reveals D's outcome before the lottery choice. Participants are incentivized to report how much they think receiving this additional information would increase their chance of receiving \$3. This question essentially elicits their information valuations. In the experiment, participants answer this question in a variety of scenarios for different values of d and a . This allows me to study how information valuations vary with the choice options in a within-subject design.

The information valuation question in this experiment has a straightforward rational answer that does not depend on risk preferences or belief-updating rules. If the information reveals that D will win, then participants should choose D, guaranteeing a \$3 win. If, instead, the information indicates D will fail, then participants should choose A, which has an a chance of winning. Essentially, if participants react to information correctly, the information induces a compound lottery with a total $d + (1 - d)a$ chance of winning. Therefore, the information increases the chance of receiving \$3 by $(1 - d)a$. This expression is easy to interpret through the lens of equation (1): the information steers the choice away from the default with a probability of $1 - d$, and when this happens, the winning chance increases from 0 to a . It also succinctly illustrates the two rational comparative statics: information valuation should decrease in d but increase in a .

I elicit this valuation in three ways, all incentivized by Becker-DeGroot-Marschak-type mech-

anisms. In the Comparison treatment, participants choose between receiving the information and increasing both lotteries' winning chances by x percentage points; their switching point x is interpreted as their information valuation. In the Direct treatment, participants simply complete the sentence: "If I find out Lottery D's outcome before choosing, my chance of choosing a winning lottery goes up by ____." In the willingness-to-pay (WTP) treatment, participants report how many "chips" they would pay for the information, where paying chips lowers the winning chance of a separate side lottery. These methods differ in whether the valuation problem is framed as a choice, a direct probability judgment, or a purchase decision, allowing me to test which patterns are robust and which are elicitation-specific. In addition, participants also answer an open-ended question about when information is most useful. This allows me to supplement behavioral data with their direct reasoning.

Across the three treatments, information valuations deviate sharply from the rational benchmark, but the form of the deviation depends on the elicitation method. In the Comparison treatment, average valuations decrease in d but remain insensitive to a . The average insensitivity to a masks substantial heterogeneity, with some participants responding in the rational direction and others in the opposite direction. The seemingly rational sensitivity to d is also driven by non-rational factors. One such factor is a satisficing heuristic, where participants choose the information unless the percentage increase raises D's winning chance above a target level. Another factor is the cap on the possible percentage increase x , which is lower in the high- d scenario than in the others, mechanically compressing valuations when d is high. Together, these forces generate a negative relationship between valuations and d even when participants are not correctly evaluating the value of information.

In the Direct treatment, where the comparison with a percentage increase and the differential answer cap are absent, many participants instead substitute the winning-chance gap $d - a$ for the value of information, producing wrong-signed average responses to both d and a . In the WTP treatment, valuations are compressed toward zero across scenarios, consistent with loss aversion, which dampens sensitivity to both parameters. These elicitation-dependent patterns suggest that

many participants rely on context-specific heuristics that often lead to deviations from rational information values. At the same time, the share of participants who respond to a in the rational direction is fairly stable across treatments, suggesting that correct understanding is less affected by elicitation details.

One possible explanation for the incorrect response to a is that, in the D-Info experiment, the information concerns D's outcome rather than A's, potentially making d more salient. In addition, in the rational value $(1 - d)a$, a governs the gain from switching conditional on D failing, which may be less direct than $1 - d$, the probability that this contingency occurs. To test these explanations, I introduce the A-Info treatment, where participants evaluate information that reveals A's outcome. This treatment has the same rational benchmark but makes A the information target and makes a the probability that information changes the choice. The treatment improves valuation in the simple scenario $a = 0$, where information is worthless, but has little effect in the more complex scenarios where information has positive value. Thus, salience and directness of the role of the alternative option cannot explain the main failure to correctly account for a .

Conceptually, even if participants attend to the relevant parameters, correctly evaluating information may still require three steps: identifying the relevant contingencies with and without information, deducing the optimal choice in each contingency, and integrating the resulting payoffs to derive the value of information. To identify which steps create difficulty, I conduct a strategy-method version of the D-Info experiment in which participants, before evaluating the information, first state which lottery they would choose without information and after each possible realization of information. All participants make the optimal lottery choice in each contingency, indicating that contingent choice itself is not difficult once the contingencies are made explicit. At the same time, this treatment modestly improves the sensitivity of information valuations to a , suggesting that some participants fail to spontaneously identify or use the relevant contingencies when valuing information. Finally, the fact that valuations remain far from the rational benchmark suggests that much of the remaining difficulty lies in integrating the payoffs implied by the contingent plan.

Payoff integration often involves complex computation, which may contribute to its difficulty.

In the main treatments, the relevant computational complexity primarily comes from reducing compound probabilities, i.e., calculating $(1 - d)a$. To test whether this complexity is central to the biases in information valuation, I design a variant of the D-Info experiment in which the two lotteries are mutually exclusive, so they cannot both win. The correct information value in this treatment is simply a , requiring minimal computation. Nevertheless, information valuation remains largely insensitive to a , suggesting that merely simplifying the required computation does little to improve valuation. Moreover, although the correct information value in this treatment does not depend on d , many participants still value information less as the default becomes more likely to win. This echoes the earlier result that the negative relationship between information valuation and the default option's winning chance, though seemingly rational, is largely a coincidence produced by the use of heuristics.

Besides computational complexity, the difficulty of payoff integration may also stem from the fact that these payoffs originate from multiple choices. This is a general property of information valuation because any valuable information must affect choice with some probability. To test this hypothesis, I design another treatment where instead of receiving information on a lottery's outcome, participants may "insure" Lottery D by using Lottery A as a back-up. With the insurance option, participants choosing D can still win \$3 if D fails, so long as A wins. Having the insurance induces a compound lottery identical to the one resulting from receiving information about D's outcome, thus their values should be the same. But unlike information, having the insurance does not change the optimal decision — whether insured or not, participants should choose D. Therefore, the valuation of insurance does not require integrating payoffs from multiple choices. In this treatment, participants' insurance valuations increase significantly as A becomes more likely to win. This sharply contrasts with the insensitivity of information valuation when A's winning chance changes. In a separate treatment, I rule out framing as an explanation for the difference between information and insurance valuations. Therefore, this result supports the hypothesis that integrating payoffs from multiple choices indeed contributes to the difficulty of information valuation.

The preceding diagnostic treatments isolate one difficulty at a time, but they do not show how close participants can get to rational valuation when several barriers are removed simultaneously. I therefore combine multiple aids in several “best-shot” treatments. In the best-shot versions of the D-Info and A-Info experiments, participants face mutually exclusive lotteries, state their contingent choices before evaluating information, and then directly report how much the information increases their chance of choosing a winning lottery. Even in this environment, which simultaneously simplifies computation, contingent planning, and elicitation, most participants do not give rational valuations. By contrast, when the same aids are combined with insurance valuation, which does not require integrating payoffs from multiple choices, a majority of valuations are exactly correct. This result suggests that the difficulties identified in this paper jointly account for a substantial share of valuation mistakes. At the same time, the contrast with the best-shot information treatments reinforces the conclusion that integrating payoffs across possible choices is a central source of difficulty in information valuation.

This paper contributes to the literature on the demand for information with instrumental value. Several studies investigate how people choose or evaluate noisy information structures (e.g., Ambuehl and Li, 2018; Ambuehl, 2021; Charness, Oprea, and Yuksel, 2021). In a closely-related concurrent paper, Guan et al. (2025) study preferences between partitional information structures when participants face a prediction task. This approach is complementary to mine: they fix the decision problem and vary the information structure, whereas I focus on fully-revealing information and vary the decision problem. Using machine learning methods, Guan et al. (2025) identify four structural characteristics that explain preferences beyond instrumental value. Several of these characteristics, such as the number of distinct posteriors and the number of signals retaining uncertainty, indicate an aversion to contingent planning and complex payoff integration, which is consistent with my findings.

Also related is the sequential search literature. For instance, Caplin, Dean, and Martin (2011) find that people stop searching once the status quo is good enough, which reflects a satisficing heuristic (Simon, 1955) similar to what I find in my experiment. While their finding is consis-

tent with rationality, my design explicitly specifies the distribution of the options' values, allowing me to detect departures from rationality. Similarly, rational inattention experiments, like Dewan and Neligh (2020), study perceptual tasks with varying stakes, mainly to measure attention costs assuming that participants understand the value of attention. In a field experiment, Bartoš et al. (2016) studies how recruiters and landlords allocate attention across applicants of different ethnicities. My experiment operates in a more controlled and abstract environment, but the results may have implications for field settings like theirs.

Moreover, a large literature studies the non-instrumental value of information (e.g., Nielsen, 2020; Masatlioglu, Orhun, and Raymond, 2021; Falk and Zimmermann, 2022; Golman, Loewenstein, Molnar, and Saccardo, 2022). These papers show that people sometimes have preferences over the quantity and timing of information, even when these factors do not affect decisions. My study controls for these non-instrumental factors so that they do not confound the interpretation of the results.

The results on mechanisms in this paper relate to three behavioral economics literatures: imperfect foresight/planning, evaluation of compound lotteries, and contingent reasoning failures. First, information valuation requires people to plan what they will choose after the information realizes. Consistent with evidence of imperfect foresight and planning (Binmore et al., 2002; Johnson et al., 2002; Martínez-Marquina and Rasocha, 2025; Chakraborty and Kendall, 2026), this paper shows that information valuations improve when people think through their future choices first. Second, information valuation often involves reducing compound probabilities, a task found to be challenging for many (Halevy, 2007; Chew, Miao, and Zhong, 2017; de Clippel, Moscariello, Ortoleva, and Rozen, 2024). While non-reduction of compound lottery may play a role in this paper's results, I show that participants do not correctly evaluate information even when no compound probabilities are involved. Third, to evaluate information, people need to integrate payoffs from different choices. Prior research has shown that people often make mistakes in decisions that require contingent thinking (Esponda and Vespa, 2014, 2021; Martínez-Marquina, Niederle, and Vespa, 2019). This paper offers a fresh insight into this literature: integrating payoffs from multiple choices is

more challenging than integrating multiple payoffs from a single choice.

2 Evidence on information demand

This section presents the design and results of the main experimental treatments, the D-Info treatments.¹ The decision environment and the rational benchmark are common across treatments, and I describe them first. I then detail the treatment variations, which use different elicitation methods and cover different parameter sets (all treatments reported in this paper are summarized in Table 1). Finally, I present and compare the results across treatments.

2.1 Common design

Participants are presented with a series of scenarios in random order. In each scenario, they consider a choice between two independent binary lotteries, D and A (referred to as “Investment A” and “Investment B”), whose winning chances d and a are known. The set of (d, a) scenarios differs across treatments, which will be detailed later. The chosen lottery yields a \$3 prize if it wins and \$0 otherwise. Lottery D is more likely to win, so it is the default option that participants should choose without additional information. Lottery A is the alternative option that is inferior to D *ex-ante*.

Before making the lottery choice, participants have the option to find out D’s outcome in advance, framed as “receiving a report” about the lottery. The value of this information, measured by how much it increases the chance of choosing a winning lottery, is what participants are asked to assess – the exact mechanism for eliciting this valuation differs across treatments, which will be detailed later. After a participant reports her valuation for every scenario, one scenario is selected at random, under which the participant makes a lottery choice. Whether D’s outcome is revealed before the lottery choice depends on the participant’s reported information valuation in that scenario via a Becker-DeGroot-Marschak mechanism (Becker et al., 1964). The participant receives

¹The interface of all experiments documented in this paper can be found at http://yuchengliang.com/brid/Instructions_BRID.pdf.

Table 1: Summary of experimental treatments

Experiment	Elicitation method	(d, a) pairs	Rational benchmark	N
D-Info (comparison)	Comparison	<i>baseline</i> : (60, 10), (60, 30), (60, 50), (50, 40), (70, 40), (90, 40) <i>simple</i> : (60, 0)	$(1 - d) a$	225
D-Info (direct)	Direct	<i>baseline</i> : (40, 20), (40, 30), (50, 20), (50, 30), (50, 40), (60, 30), (60, 40), (60, 50), (70, 40), (70, 50), (70, 60), (80, 50), (80, 60) <i>simple</i> : (60, 0), (100, 40)	$(1 - d) a$	104
D-Info (WTP)	WTP	the baseline pairs in D-Info (comparison)	$(1 - d) a$	70
A-Info	Comparison	same as D-Info (comparison)	$(1 - d) a$	224
D-Info (strategy method)	Comparison	the baseline pairs in D-Info (comparison)	$(1 - d) a$	73
D-Info (mutually exclusive)	Comparison	(40, 10), (40, 20), (40, 30), (30, 20), (50, 20)	a	74
D-Insured	Comparison	the baseline pairs in D-Info (comparison)	$(1 - d) a$	142
A-Insured	Comparison	the baseline pairs in D-Info (comparison)	$(1 - d) a$	156
D-Info (best-shot)	Direct	(20, 10), (30, 10), (30, 20), (40, 10), (40, 20), (40, 30), (50, 10), (50, 20), (50, 30), (50, 40)	a	25
A-Info (best-shot)	Direct	same as D-Info (best-shot)	a	22
D-Insured (best-shot)	Direct	same as D-Info (best-shot)	a	45

Notes: This table summarizes all experimental treatments reported in the main text. The Comparison elicitation method asks participants to compare the value of information against an increase in both lotteries' winning chances; the Direct method asks them to directly state the increase in winning probability that information/insurance would bring; and the WTP method asks them to pay chips (winning probability of a side lottery) for information. Each (d, a) pair lists the percentage chances of D and A winning. Simple scenarios are those where the rational information value is exactly zero ($d = 100$ or $a = 0$); the rest are baseline scenarios.

the realized outcome of her chosen lottery as her bonus payment.

After the lottery choice but prior to revealing the outcomes, participants are asked to describe in words the kinds of scenarios in which information about D's outcome is more valuable. This open-ended question is incentivized in some treatments, which will be detailed later. In an end-

line questionnaire, I collect sociodemographic information and ask unincentivized questions about participants' tendencies to gather information, plan, and take risks in their daily life (see Table A1 for descriptive statistics).

The experiment was pre-registered and conducted on Prolific with a \$2 participation fee. Participants receive extensive instructions on the details of the tasks and need to correctly answer several comprehension questions before proceeding. A total of 1246 participants were recruited across all treatments, with the experiment's median duration being 10 minutes.

2.2 Rational benchmark

The information valuation question has a rational answer. Without knowing D's outcome in advance, participants should choose D, which has a winning chance of d . If D's outcome is revealed before the lottery choice, participants should choose D if it wins and A if it does not. This strategy induces a compound lottery as illustrated in Figure 1, resulting in a total winning chance of $d + (1 - d)a$. Therefore, learning D's outcome should increase the winning probability by $(1 - d)a$. Importantly, this expression is unaffected by risk preferences and any deviations from Bayes' rule. It also has a straightforward interpretation: the information diverts the choice from the default option with probability $1 - d$, and when this happens, the winning chance increases from 0 to a . The rational information value decreases in d and increases in a .

The expression $(1 - d)a$ considers only the instrumental value of information, but prior research has shown that people may have preferences over the amount and timing of uncertainty resolution for non-instrumental reasons. In the experiment, the amount and timing of uncertainty resolution are carefully controlled, so non-instrumental factors should have minimal impact on information valuations. First, the outcomes of both lotteries will eventually be revealed irrespective of the participants' valuations, so curiosity is held constant across responses. Second, although a high reported valuation may cause D's outcome to be revealed before A's, the time gap between the two revelations is short, and intrinsic preferences for information timing are typically weak in such scenarios (Nielsen, 2020).

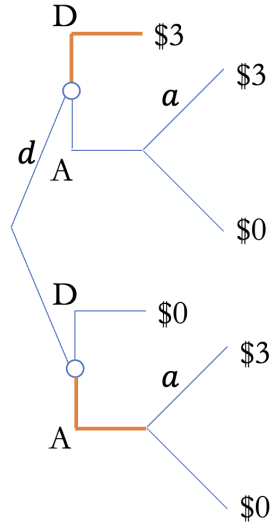


Figure 1: Payoff structure induced by the information

Notes: This figure depicts the payoff structure induced by the information about Lottery D's outcome. The straight branches represent uncertainty resolution. The letter on a straight branch represents its probability. The elbow branches are choice options. The optimal choices, shown in orange, induce a compound lottery.

2.3 Treatment variations

Within the common experimental design framework, I implement three treatments using different elicitation methods for information valuation, each with distinct advantages and limitations. These treatments also differ in the set of (d, a) scenarios covered.

2.3.1 Comparison treatment ($N = 225$)

In this treatment, all participants evaluate information in 6 baseline scenarios presented in random order, each scenario with a different (d, a) pair: $(60, 10)$, $(60, 30)$, $(60, 50)$, $(50, 40)$, $(70, 40)$, and $(90, 40)$, all in percentage-point terms. This set of parameters includes variations in a holding d fixed and variations in d holding a fixed, which allow me to test the comparative statics predictions of the rational benchmark.

Among these participants, 78 of them also face a scenario with $(d, a) = (60, 0)$ presented before all baseline scenarios. Information valuation is arguably simpler in this scenario, as all

it takes for participants to correctly evaluate information at zero is to recognize that the optimal lottery choice does not depend on the information realization. I can test whether information valuation in this scenario exhibits distinct patterns from the rest.

Information valuation is elicited through comparisons between (a) the information about D's outcome (referred to as "the report" in the experiment) and (b) an x -percentage-point increase in both lotteries' winning chances (referred to as "an $x\%$ -upgrade").² Specifically, participants select one from a list of statements of the form

I would choose the report over an $(x - 1)\%$ -upgrade, but I would choose an $(x + 1)\%$ -upgrade over the report.

From the top of the choice list to the bottom, the number x increases from 0 to 30 in steps of 2, except for the scenario with $d = 90$ where the maximum x is 10.³ The selected x is interpreted as the participant's information valuation. The Becker-DeGroot-Marschak incentive scheme ensures truthful reporting: a random number y is drawn after the valuation, and the participant receives the information if her valuation is at least y and the increase in winning chances of $y\%$ otherwise.⁴

The open-ended question about how to value information is also incentivized. Specifically, participants are asked to provide advice on this to other participants, and if the advisee chooses a winning lottery, the advisor will receive an additional \$2 bonus as well.⁵

The Comparison treatment has the advantage of rendering not only risk preferences, but also loss preferences and probability weighting irrelevant. Because participants compare two probability increments from the same status quo d , reference dependence and probability weighting do not affect responses. Moreover, increasing both d and a rather than only d does not alter the relative salience of the two lotteries. The main limitation is that the maximum x is mechanically capped at

²In all treatments, I use the "%" symbol to represent percentage points to avoid unfamiliar jargon. The experiment includes a comprehension question about this to ensure that participants understand this representation correctly.

³The first part of the statement is omitted when $x = 0$, and the second part when $x = 30$. This elicitation format is similar to the sMPL in Andersen et al. (2006), which directly asks participants to state a switching point in a multiple price list.

⁴Following Danz, Vesterlund, and Wilson (2022), the instructions simply state that it is in the participants' best interest to answer the questions based on their true preferences. The details of the incentive scheme are described in the instructions, but participants are not required to read them.

⁵Results on the advisees' behavior are reported in Appendix D.1.

10 when $d = 90$, which is different from the other scenarios where x can reach 30. In addition, the comparison statement may appear unusual to some participants.

2.3.2 Direct treatment ($N = 104$)

This treatment includes 13 randomly-ordered baseline scenarios with the following (d, a) pairs: $(40, 20)$, $(40, 30)$, $(50, 20)$, $(50, 30)$, $(50, 40)$, $(60, 30)$, $(60, 40)$, $(60, 50)$, $(70, 40)$, $(70, 50)$, $(70, 60)$, $(80, 50)$, and $(80, 60)$. In addition, participants face 2 simple scenarios, $(60, 0)$ and $(100, 40)$, presented either at the start or at the end.⁶ This allows me to test the rational comparative statics across a wider set of parameters.

Participants directly report their assessment of how much the information would raise their winning probability by completing the sentence

If I find out [Lottery D]’s outcome before choosing, my chance of choosing a [winning lottery] goes up by ____.⁷

Across all scenarios, the available responses are 0%, 1%, . . . , 29%, and “30% or more.” The sentence itself is phrased as a pure probability judgment problem without mentioning any information choice. Nevertheless, the mechanics of the underlying incentive scheme are identical to those in the Comparison treatment, and participants can reveal the details by clicking a button on the instructions screen.⁸

The open-ended question about how to value information is not incentivized in this treatment. Participants are simply asked the following question: “In what kinds of scenarios would finding out [Lottery D]’s outcome most increase your chances of choosing a [winning lottery]?”

The Direct treatment has the advantage of maintaining a consistent answer range across scenarios, removing the cap that affects the $d = 90$ scenario in the Comparison treatment. Moreover, the

⁶The scenario $(100, 40)$ is simple because in order to correctly evaluate the information about D at zero, participants only need to recognize that it has only one possible realization.

⁷“Lottery D” is referred to as “Investment A,” and “winning lottery” is referred to as “successful investment.”

⁸This design is again informed by Danz et al. (2022) who show that separating the cognitive task from the incentive mechanics can reduce participants’ non-optimal strategizing behaviors.

information valuation question is arguably easier to understand, as it does not involve any comparison object. However, by framing the question as a pure belief-elicitation problem, the elicitation may trigger reasoning that differs from how participants think when they face a problem framed directly as a choice.

2.3.3 WTP treatment ($N = 70$)

This treatment covers the same six baseline scenarios in the Comparison treatment and no simple scenarios.

Participants pay “chips” to buy the information about D’s outcome, where each chip paid reduces the winning chance of a separate side lottery X by one percentage point. With probability 50%, the final payment is determined by the chosen lottery between D and A, mirroring the Comparison treatment; otherwise, the payment is determined by X , whose percentage winning chance equals 60, the endowment, minus the number of chips the participant pays. In each scenario, participants report how many chips they are willing to pay to receive the information by selecting one from a list of statements of the form

I would buy the report if it costs $x - 1$ chips, but not if it costs $x + 1$ chips,

where x increases from 0 to 30 in steps of 2 for all scenarios.⁹ The x in the selected statement is interpreted as the participant’s information valuation, and the BDM mechanism is used to incentivize the answers. Participants also write incentivized advice as in the Comparison treatment.

The WTP treatment has the advantage of maintaining a consistent answer range across scenarios, similar to the Direct treatment. In addition, the buy-frame may also be easier for some participants to understand, as prior research finds that people are sometimes more rational and forward-looking when buying than when evaluating in other ways (Grosshans et al., 2020; Akepanidaworn et al., 2023). The treatment introduces two new drawbacks. First, the side lottery X adds complexity. Second, the WTP treatment may be affected by loss aversion: participants may be reluctant to

⁹The first part of the statement is omitted when $x = 0$, and the second part when $x = 30$.

give up chips and therefore underreport their valuations across scenarios. More importantly, this compression of WTP in level may also attenuate its sensitivity to the parameters.

2.4 Aggregate results

Figure 2 presents the average information valuations across the three treatments. Each panel plots the mean valuation against the rational benchmark $(1 - d)a$, with one dot per (d, a) scenario. Solid lines connect scenarios with the same d (varying a); dashed lines connect scenarios with the same a (varying d). The 45-degree gray line represents the rational benchmark.¹⁰

The three elicitation methods produce qualitatively different patterns, but all three depart from the rational benchmark.

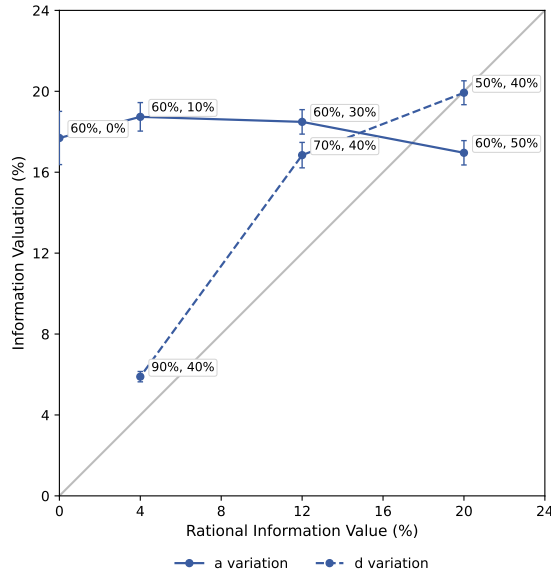
In the Comparison treatment, valuations decrease in d , consistent with the rational benchmark. Along the dashed line in Figure 2a, valuations fall sharply from 19% at $d = 50\%$ to 5% at $d = 90\%$, mirroring the rational decline from 20% to 4%. Along the solid lines, however, valuations are flat or even slightly decreasing in a , contradicting rationality. At the simple scenario ($d = 60, a = 0$), the average valuation is 17.7% — lower than the average valuation at the closest baseline scenario ($d = 60, a = 10$). The simple scenario thus reverses the wrong-direction pattern observed when a varies along the baseline scenarios, but the valuation is still far above the rational value of zero.¹¹

In the Direct treatment (shown in Figure 2b), valuations are sensitive to both d and a in the *wrong* direction. Along the dashed lines, valuations rise rather than fall as d increases; along the solid lines, valuations fall rather than rise as a increases. At the two simple scenarios, valuations are significantly lower than at the nearest baseline parameter values, and the share of correct answers is significantly higher: 32.7% in the $(60, 0)$ scenario and 43.3% for the $(100, 40)$ scenario compared to 8.0% for the rest. However, the average valuations remain far from the rational value of zero.

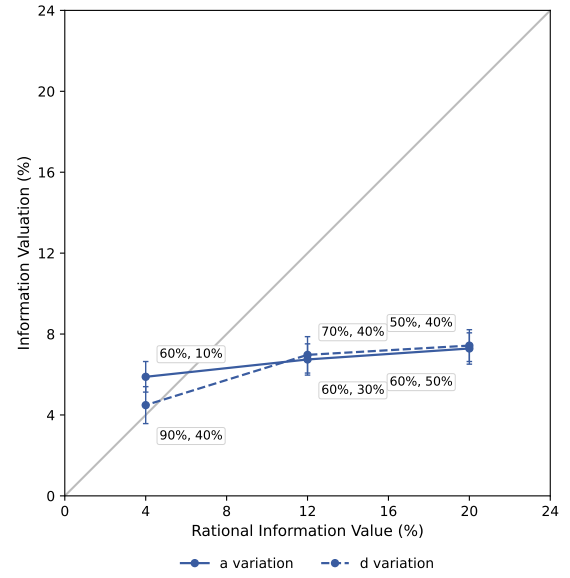
In the WTP treatment (shown in Figure 2c), valuations are significantly compressed compared

¹⁰In the Comparison treatment, whether a simple scenario is included or not does not significantly affect valuations of the baseline scenarios. In the Direct treatment, the order of the simple scenarios also does not significantly affect valuations of the complex scenarios or the simple scenarios themselves. Therefore, in both treatments, I pool data across these variations throughout.

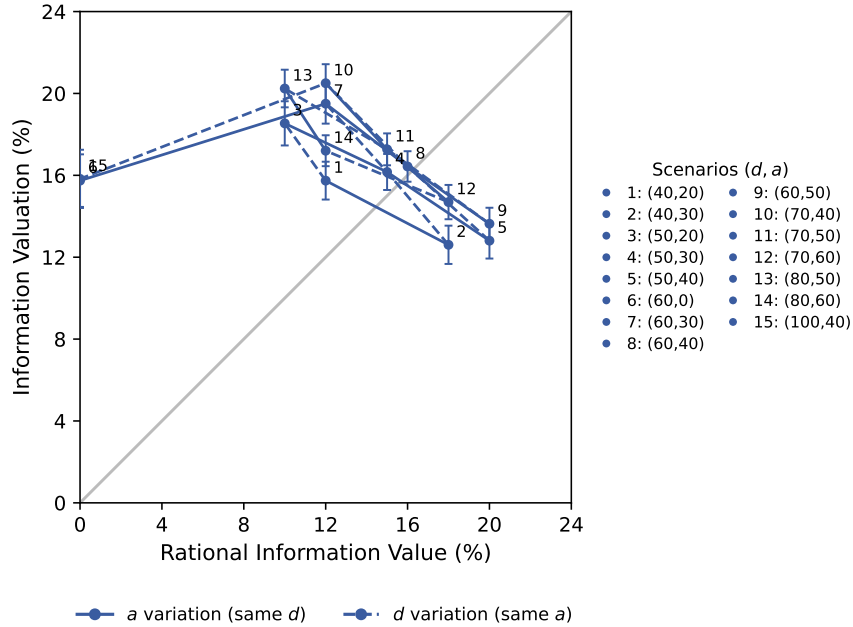
¹¹The share of correct answers in the simple scenario is 19.2%, higher than the 6.5% across the baseline scenarios.



(a) Comparison ($N = 225$)



(c) WTP ($N = 70$)



(b) Direct ($N = 104$)

Figure 2: Information valuations across the three elicitation methods

Notes: This figure shows the average information valuation in each (d, a) scenario. Each dot represents one (d, a) scenario. In panels (a) and (c), each dot is labeled by its (d, a) values; in panel (b), the side legend numbers each scenario from 1 to 15 to keep the plot uncluttered. Solid lines connect scenarios with the same d (varying a); dashed lines connect scenarios with the same a (varying d). The 45-degree gray line represents the rational benchmark $(1 - d)a$. Error bars represent ± 1 standard error.

to the Comparison treatment. Despite the compression, valuations vary mildly across scenarios in roughly the rational directions — decreasing in d along the dashed lines and increasing in a along the solid lines.

To quantify these visual patterns, I estimate the within-treatment regression

$$V_{is} = \alpha_i + \beta_d \cdot d_s + \beta_a \cdot a_s + \varepsilon_{is} \quad (2)$$

separately for each treatment, where V_{is} is participant i 's reported valuation in scenario s , d_s and a_s are the winning probabilities of D and A, and α_i is a participant fixed effect. Simple scenarios ($d = 100$ or $a = 0$) are excluded from all three regressions. Standard errors are clustered at the participant level.¹²

Table A2 reports the results, which confirm the visual patterns. In the Comparison treatment, $\beta_d = -0.349$ ($p < 0.001$) is correctly signed and large, while $\beta_a = -0.043$ ($p = 0.005$) is wrong-signed and small in magnitude. In the Direct treatment, both estimated sensitivities are directionally opposite to the rational benchmark: $\beta_d = +0.340$ and $\beta_a = -0.286$ (both $p < 0.001$). In the WTP treatment, both are correctly signed but compressed: $\beta_d = -0.073$ ($p = 0.002$) and $\beta_a = +0.036$ ($p = 0.042$). The sensitivity to a is numerically about half of that to d , but the difference in magnitude is imprecisely estimated.

2.5 Individual-level distributions of sensitivities

To understand why average information valuation deviates from the rational benchmark but exhibits markedly different patterns across the three elicitation methods, I investigate the results on the individual level. Specifically, I run the following regression separately for each participant

¹²Adding the interaction term $d \cdot a$ to this specification would let it nest the rational benchmark, since $(1 - d)a = a - d \cdot a$. However, in both the Comparison treatment and the WTP treatment, $d \cdot a$ is exactly collinear with d and a over the six baseline scenarios, so its coefficient cannot be identified from the within-subject variation. In the Direct treatment, where the parameter set is richer, the interaction is identified, but its coefficient is small in magnitude (-0.0003 , standard error 0.0010) and statistically indistinguishable from zero ($p = 0.748$). I therefore report the linear specification throughout the main text.

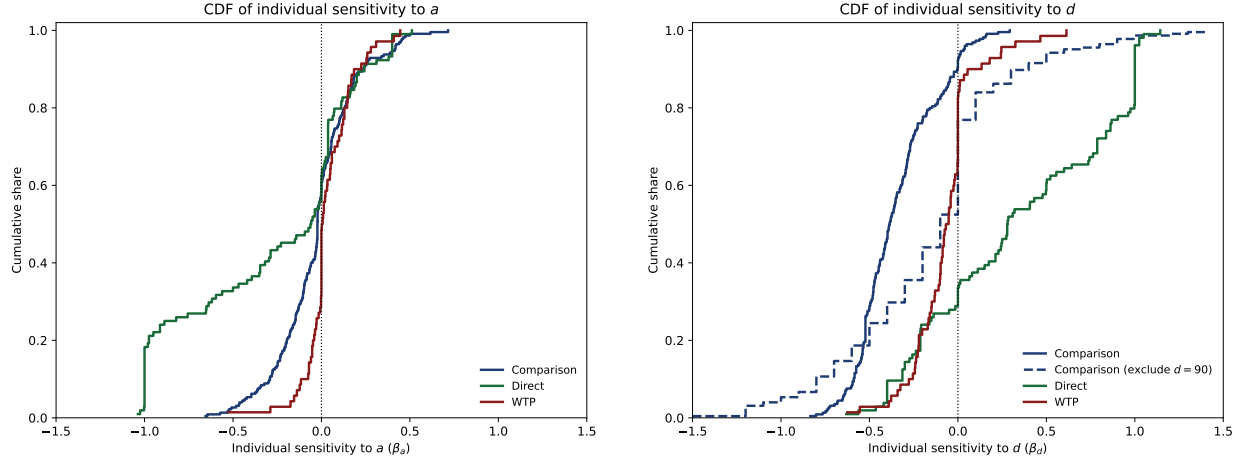


Figure 3: CDF of individual sensitivity to a and d across elicitation methods

Notes: This figure shows the empirical cumulative distribution functions of individual sensitivities β_a^i (left) and β_d^i (right), estimated from the participant-level regression in equation (3). Sensitivities are clipped to $[-1.5, 1.5]$ for visualization. The vertical dashed line at zero marks the boundary between correct-direction and wrong-direction responses (correct: $\beta_a^i > 0$ on the left, $\beta_d^i < 0$ on the right). All regressions exclude simple scenarios with $d = 100$ or $a = 0$. The right panel reports two versions of β_d^i for the Comparison treatment: the solid blue line uses all baseline scenarios, while the dashed blue line exclude the scenario with $d = 90$ (where the answer cap is fixed at 10%, in contrast to the 30% cap that applies to the other scenarios).

i :

$$V_{is} = \alpha_i + \beta_d^i \cdot d_s + \beta_a^i \cdot a_s + \varepsilon_{is}. \quad (3)$$

The coefficients β_d^i and β_a^i summarize how participant i 's valuations respond to changes in each parameter. Under rationality, $\beta_d^i < 0$ and $\beta_a^i > 0$. For all treatments, I exclude simple scenarios where $d = 100$ or $a = 0$.

Figure 3 plots the empirical cumulative distribution function (CDF) of β_a^i (left panel) and β_d^i (right panel) across the three treatments.

Sensitivity to a (left panel). The distribution of participants who respond to a in the correct (positive) direction — the upper portion of the CDF — looks remarkably similar across the three elicitation methods. This is confirmed by pairwise Kolmogorov-Smirnov tests, which cannot reject the null that the distributions conditional on $\beta_a^i > 0$ are identical (all $p > 0.21$). This result suggests that the share of participants who correctly understand the directional impact of A on information

valuation is stable across elicitation methods.

By contrast, the rest of the distribution, where participants are insensitive to a or respond in the wrong direction, differs substantially across treatments. The WTP distribution is the most compressed toward zero, with a pronounced mass of participants very close to $\beta_a^i = 0$. This compression mirrors the aggregate result and is consistent with loss aversion, as participants who are reluctant to give up chips will report low valuations for information, limiting their sensitivity to the parameters. The Direct treatment, by contrast, exhibits a mass of participants near $\beta_a^i = -1$, which is a unique pattern with no analog in the other two treatments.

Sensitivity to d (right panel). The distribution of β_d^i also differs across the three treatments. First, the share of participants who react to d in the correct (negative) direction is much larger in the Comparison treatment than in the other two treatments. However, this is in large part due to the low cap on information valuation answers when $d = 90$; if we restrict attention to scenarios where the answer cap is consistently 30% (the dashed line), the excess mass of d -correct responses in the Comparison treatment shrinks substantially and is no longer larger than in the WTP treatment.

Second, the WTP treatment again displays a mass near zero, contributing to the flattening of the average sensitivity visible in Figure 2.

Third, the Direct treatment displays a mass near $\beta_d^i = +1$, again the opposite of the correct direction.

Are reactions to d and a correlated? To study the joint distribution of directional reactions to the two parameters, I first classify each individual-level regression coefficient into three bins: decreasing $(-\infty, -0.05)$, near-constant $[-0.05, 0.05]$, and increasing $(0.05, \infty)$. The per-treatment shares are reported in Table A3. I then cross-tabulate the classifications of β_d^i and β_a^i and calculate, for each treatment, the Spearman correlation between $-\beta_d^i$ and β_a^i in Table A4.

Correct directional responses to d and a are significantly positively correlated in the Comparison treatment ($\rho = +0.22$, $p = 0.001$). This correlation remains if β_d^i is estimated excluding the $d = 90\%$ scenario, where information valuations are capped at 10%. In the Direct treatment, the

correlation is largest ($\rho = +0.81$, $p < 0.001$), reflecting both a large number of participants who get both directional responses wrong and a substantial share who get both right. The correlation in the WTP treatment is positive but smaller and imprecisely estimated ($\rho = +0.07$, $p = 0.59$), which partly reflects the large share of participants who are close to insensitive to both parameters.

The gap heuristic in the Direct treatment. In the Direct treatment, the mass at $\beta_a^i \approx -1$ and $\beta_d^i \approx +1$ as well as the high association between wrong directional responses to these two parameters suggests that many participants may be using a heuristic based on $d - a$ to evaluate the information. Indeed, 38.5% of all responses in the 13 baseline scenarios are exactly equal to $d - a$. The prevalence of this heuristic may be explained by attribute substitution theory (Kahneman and Frederick, 2002; Fan et al., forthcoming): because it is difficult to answer how much revealing D’s outcome increases the winning chance, people answer how much higher D’s winning chance is than A’s — an easier and superficially similar question.¹³

2.6 Evidence from participants’ own reasoning

The behavioral patterns above suggest that participants in the three treatments use different decision rules to evaluate the same information. To pin these decision rules down, I examine participants’ open-ended responses about when information is more valuable. I classify these responses into themes using a large language model (Claude Opus 4.6): I first ask the model to inductively propose themes from random samples of 50 responses per treatment (three independent runs per treatment for stability), then consolidate the discovered themes into a single codebook spanning all treatments, and finally apply the codebook to every response one at a time with deterministic settings. The full prompts and codebook are reported in Appendix B. Table 2 lists eight select themes which are particularly informative about participants’ reasoning and reports the share of participants in each treatment who express each theme.

¹³One may worry that this answer pattern instead reflects confusion about what is being elicited. I include a comprehension check to make sure that this is not the case. In addition, in Section 3.4, I present a treatment that achieves a substantial level of rationality using the same elicitation method.

Table 2: Prevalence of reasoning themes by treatment

% of participants whose reasoning expresses...	Comparison	Direct	WTP
High $d \Rightarrow$ info more valuable (wrong direction on d)	8.6	17.3	1.4
High $a \Rightarrow$ info more valuable (correct direction on a)	8.6	25.0	2.9
Switch to A if D fails (correct contingent plan)	5.0	8.7	0.0
Satisficing on post-increase winning chance	13.1	0.0	0.0
Satisficing on prior winning chance (info preferred when d is low)	3.2	1.0	14.5
Large $d - a \Rightarrow$ info more valuable	4.5	21.2	1.4
Small $d - a \Rightarrow$ info more valuable (closer to correct)	6.8	9.6	10.1
Cost concern (loss aversion)	0.9	0.0	46.4

Notes: Each cell is the share of participants whose open-ended advice or reasoning expresses the theme, as classified by Claude Opus 4.6. Themes are non-exclusive: a single response can be coded into multiple themes. “Satisficing on post-increase winning chance” captures responses with a cutoff on D’s *post-upgrade* chance (e.g., “I take the increase if it pushes D above 70%”); “Satisficing on prior winning chance” captures responses with a cutoff on D’s *prior* chance, with the report preferred when D is uncertain or low (e.g., “the report matters more when D is around 50%”). The table shows 4 of the 12 master codebook themes plus 4 sub-themes; the remaining 8 themes (e.g., “mentions the spread between D and A” without a directional claim, “always prefer the report”, “gut feeling”) are less informative about participants’ reasoning and are omitted here for compactness. The full prevalence of all 12 themes by treatment is reported in Appendix Table A9. See Appendix B for the codebook and prompts.

The themes extracted from the coding exercise are consistent with the behavioral patterns documented above. For example, the gap heuristic — articulated as “information is more valuable when the spread between D and A is large” — is prevalent (21%) only in the Direct treatment. This treatment also has the highest share (17%) of participants who explicitly say information is more valuable when D’s probability is high. This mirrors the wrong-direction sensitivity to d and the prevalence of the gap heuristic in the behavioral data. In contrast, many participants in the WTP treatment (46%) mention the loss aversion theme, which echoes the compressed sensitivity to d and a in the behavioral data.

Interestingly, even among the reasoning types that generate the correct comparative statics on

d , not all reflect rational reasoning about information values. Specifically, 13.1% of Comparison treatment participants describe satisficing decision rules that set targets for the post-increase winning chances: they prefer the increase to the information whenever it can push D’s chance up to that target.¹⁴ A participant who follows such a rule will appear to value the information more highly when the pre-increase d is lower, producing a negative β_d , but this pattern is mechanical rather than driven by reasoning about information value. This reasoning type is absent in the Direct treatment because the elicitation there is framed as a pure judgment problem and does not present a choice between the information and an increase in the lotteries’ winning chances. The differential prevalence of this satisficing heuristic helps explain why sensitivity to d appears closer to rational in the Comparison treatment than in the Direct treatment.¹⁵

Finally, the Direct treatment contains relatively more correct reasoning about the role of a : 8.7% of participants mention switching to A if D fails, and 25% mention that the information is more valuable when a is higher. These shares are higher than in the other two treatments, which may reflect the simplicity of asking participants to state the effect of information directly.

To examine whether participants’ open-ended explanations are consistent with their own information valuations at the individual level, I conduct another classification of the explanations that is more comparable with the classification of behavioral responses. Specifically, I use the same large language model to classify whether each response mentions each lottery and, if so, the implied directional impact of its winning chance on the value of the information (see Table A5 for the classification result by treatment and Appendix C for the codebooks). I then cross-tabulate this classification with the classification of individual-level regression coefficients reported in Table A3.¹⁶ Tables A6 and A7 show the results. Across all three treatments and both parameters,

¹⁴Examples from the advice include “I try to at least get one of the percentages to 70% if possible,” “I would safely upgrade your chances to at least 70% for Lottery D,” and “I think it’s best to have a predetermined percentage as your target for Lottery D. For example, if 85% is my target, then I would choose the answer that makes Lottery D equal to 85% or more.”

¹⁵A similar decision rule also appears in the WTP treatment, where 14.5% of participants value the report more when d is lower. It is less clear what reasoning underlies this pattern. It could reflect either rational reasoning or satisficing on the total winning chance from the choice between D and A. Regardless of the underlying mechanism, the prevalence of this rule in WTP contributes to the negative average β_d there.

¹⁶In the cross-tabulations, responses that do not mention a lottery are pooled with those that mention the lottery without taking a directional stance.

self-written reasoning about the comparative statics is significantly and positively correlated with the behavioral patterns (Spearman ρ ranges from +0.26 to +0.44, all $p < 0.05$).¹⁷

2.7 Summary

In the main D-Info experiment, the rational value of learning D's outcome before choosing between lotteries D and A is $(1 - d)a$, which decreases in d and increases in a . Across three treatments that vary the elicitation method, the share of participants who respond to a in the correct direction is fairly stable, suggesting that correct understanding of a 's impact on information value is less affected by elicitation details. By contrast, participants who do not understand information value exhibit different patterns across treatments, producing different aggregate relationships between valuations and the two parameters.

In the Comparison treatment where participants evaluate the information against increases in the lotteries' winning chances, average valuation decreases in d but is insensitive to a . The seemingly rational sensitivity to d can be partly attributed to a satisficing heuristic that sets a target for D's winning chance after the increase. In addition, the close-to-rational information valuation when $d = 90$ is mechanically influenced by the low answer cap in that scenario. In the Direct treatment, these two features are removed, and many participants instead heuristically evaluate information at $d - a$, generating wrong-signed responses to both d and a . In the WTP treatment, valuations are compressed across scenarios, consistent with loss aversion, which dampens sensitivity to both parameters.

Taken together, information valuation depends strongly on the elicitation method for those who do not correctly understand its value: these participants appear to rely on context-specific heuristics that often lead to mistakes.

These prevalent valuation mistakes and context-dependent heuristics suggest that information valuation is difficult for many participants. The next section investigates the sources of this diffi-

¹⁷The exception is the Comparison treatment on d , where the correlation is small and insignificant in the full sample ($\rho = +0.10$, $p = 0.13$) but rises to +0.28 ($p < 0.001$) once β_d^i is re-estimated excluding the $d = 90\%$ scenario, where the differential answer cap mechanically inflates the negative- β_d^i bin. See Panels A and B of Table A6.

culty using diagnostic treatments.

3 Mechanisms

3.1 Highlighting the alternative option

In the D-Info experiment, Lottery D is the target of information, which could make d more salient than a . Moreover, in the rational value $(1 - d)a$, a governs the gain in winning probability conditional on D failing, whose effect on payoffs may be less direct than $1 - d$, the probability that this contingency occurs.¹⁸ Consistent with this asymmetry, Table A5 shows that participants are more likely to mention D than A in their open-ended explanations in the D-Info treatments. The differential prominence of D and A — both its salience as the information target and its directness in determining the value of information — could potentially explain why information valuations do not respond to a correctly for many participants.

To test this explanation, I conduct the A-Info treatment ($N = 224$), where participants evaluate information that reveals the outcome of A instead of D before the lottery choice. This design highlights A as the focus of information. Moreover, while the A-Info treatment has the same rational answer as the main experiment, it flips the roles of d and a in the formula. To see why, note that participants receiving information about A should choose A if it wins and D if not. This strategy induces the compound lottery in Figure 4a and gives a $a + (1 - a)d$ winning chance. Therefore, the value of learning A's outcome is $a(1 - d)$, the same as the value of learning D's outcome. In this formula, a is the probability that information diverts the choice from the default option and $1 - d$ is the increase in winning chances when this happens. Thus, under the direct/indirect interpretation, the A-Info treatment makes a the more direct component of information value.

The elicitation method and parameter set in the A-Info treatment are identical to those in D-Info (comparison). 72 participants face the simple scenario $(d, a) = (60, 0)$ first, whereas the rest

¹⁸Frederick et al. (2009) and Dal Bó et al. (2018) show, in different contexts, that people are less likely to internalize indirect effects.

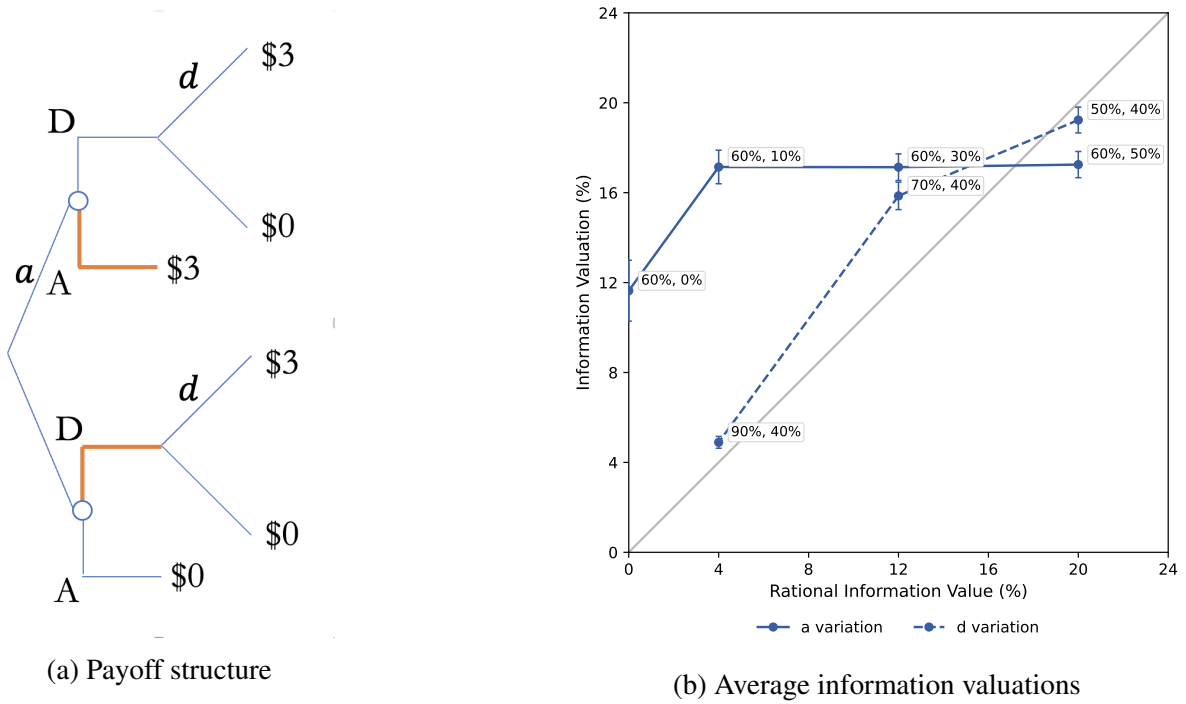


Figure 4: The A-Info treatment: payoff structure and information valuations

Notes: Panel (a) depicts the payoff structure induced by the information in the A-Info treatment. The straight branches represent uncertainty resolution; the letter on a straight branch represents its probability; the elbow branches are choice options; the optimal choices, shown in orange, induce a compound lottery. Panel (b) shows the average information valuations in the A-Info treatment ($N = 224$). Each dot represents one (d, a) scenario, with the label showing the values of d and a . Solid lines connect scenarios with the same d (varying a); dashed lines connect scenarios with the same a (varying d). The 45-degree gray line represents the rational benchmark $(1 - d)a$. Error bars represent ± 1 standard error.

only encounter the six baseline scenarios.

Table A5 shows that 42.7% of participants mention Lottery A in their open-ended explanations, higher than the 31.1% in D-Info (comparison). This indicates that making A the information target successfully increases its prominence. Despite the effective treatment manipulation, information valuation shows only limited improvement. Figure 4b plots the average information valuation by scenario: valuations decrease in d in the correct direction but are flat or slightly decreasing in a , mirroring D-Info (comparison). The visual pattern is confirmed by regression analysis. At the aggregate level, regression coefficient β_a in A-Info (-0.001) is only modestly less wrong-signed than in D-Info (comparison) (-0.043 ; see Table A2). Similarly, at the individual level, the share of A-Info participants with $\beta_a^i > 0.05$ is 37.5%, only modestly higher than the 28.4% in D-Info

(comparison).¹⁹

The improvement is clearer in the simple scenario ($d = 60, a = 0$), where the rational valuation is exactly zero. Among participants who face this scenario in the A-Info treatment, the average valuation is 11.6%, significantly lower than the 17.7% when D is the target of information. The share of participants who report exactly zero is also nearly twice as large in A-Info (37.5% vs. 19.2%, $z = 2.49, p = 0.013$).

The A-Info treatment therefore suggests that making A the information target can help participants recognize that information has no value at the transparent simple case $a = 0$. Recognizing $a = 0$ is sufficient to see that learning A's outcome cannot improve the lottery choice. In contrast, when information has positive value and valuation requires more complex contingent reasoning and calculation, changing the information target has little effect.

The next section therefore focuses on the complex scenarios and studies why participants struggle when information has positive value.

3.2 Contingent thinking

Even if participants attend to the relevant parameters, correctly evaluating information may still require three conceptual steps:

Step 1: Identifying the relevant contingencies: the no-information case and each possible realization of information.

Step 2: Deducing the optimal choice in each contingency.

Step 3: Integrating the resulting payoffs across contingencies to derive the value of information.

To identify which steps create difficulty, I conduct a strategy-method version of the D-Info experiment. In D-Info (strategy method) ($N = 73$), I elicit valuations of information about Lottery D's outcome in the six baseline scenarios from D-Info (comparison) using the same elicitation method. The key difference is that before evaluating the information, participants first report their contingent lottery choices: which lottery they will choose if they learn that D will win, which

¹⁹The per-treatment distribution of individual β^i across all 11 treatments is reported in Appendix Table A3.

lottery they will choose if they learn that D will lose, and which lottery they will choose if they do not receive information about D's outcome. After information valuation, once it is determined whether they receive information, their contingent lottery choice is implemented automatically and the outcomes are revealed.

This treatment helps separate the roles of the three conceptual steps. Prompting participants to formulate a contingent lottery-choice plan makes the relevant contingencies explicit before valuation (Step 1). It also reveals whether participants can identify the optimal lottery choice within each contingency (Step 2). If participants choose optimally, then the remaining task is to integrate the payoffs implied by the choice plan (Step 3), so remaining deviations point primarily to payoff integration.²⁰

In the strategy-method treatment, all participants make optimal contingent lottery choices, suggesting that Step 2 is not a major barrier once the contingencies are explicit. Figure 5 presents the comparative statics of information valuations. Average information valuation shows a slight but statistically insignificant increase ($p = 0.136$) as Lottery A becomes more likely to win. However, this is a significant change from D-Info (comparison) ($p = 0.008$), where average information valuation decreases in a . The improved sensitivity to a is also evident at the individual level: 38.4% of Strategy-method participants have $\beta_a^i > 0.05$, compared with 28.4% in D-Info (comparison) ($p = 0.11$).

To evaluate information, people need to integrate multiple potential payoffs, which can be difficult for at least two reasons. First, payoff integration can be computationally complex. Second, the payoffs to be integrated come from multiple choices, which could add to the representational complexity.

These findings indicate that participants have no trouble formulating an optimal contingent plan once the contingencies are made explicit. At the same time, the fact that simply prompting participants to make contingent plans improves information valuation suggests that incorrect valuation, at least partly, reflects a failure to engage in contingent thinking in the first place. The persistence of

²⁰Because all uncertainties are resolved in one shot with or without information, preferences for the timing of uncertainty resolution are also irrelevant to information valuation in this treatment.

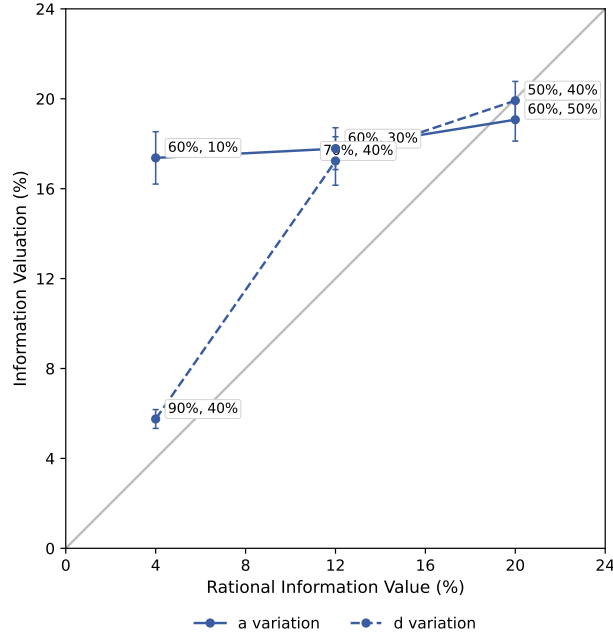


Figure 5: Information valuations in the Strategy-method treatment

Notes: This figure shows the average information valuations in the D-Info (strategy method) treatment. Each dot represents a scenario, with the tag next to it showing the d of that scenario. Error bars represent 95% confidence intervals.

substantial deviations, however, indicates that many participants still struggle to integrate payoffs even when the contingent choices are already formulated.

In the next subsection, I zoom in on the difficulty of payoff integration and investigate its sources.

3.3 Payoff integration

3.3.1 Computational complexity

In the treatments discussed so far, the computational complexity of correctly evaluating information mainly comes from the reduction of compound probabilities, i.e., multiplying $(1 - d)$ with a . To investigate whether computational complexity is key to the biases in information valuation, I design a variant of the main experiment, named D-Info (mutually exclusive) ($N = 74$), that eliminates the need for reducing compound probabilities. This is achieved by making the success of

D and A mutually exclusive: D wins with probability d , A wins with probability a , and with the remaining probability both lotteries lose; the two lotteries cannot both win. As in the other D-Info treatments, participants should choose D if the information says it wins and choose A otherwise. This strategy leads to a lottery depicted in Figure 6a. The total winning chance by following this strategy is $d + a$, which implies that the information value should be a . Note that this expression does not involve any reduction of compound probabilities, thereby significantly reducing computational complexity. It also does not involve d , which allows me to test whether information valuation responds to the default lottery when it should not. I elicit information valuations in 5 scenarios using the same method as in D-Info (comparison): the values of (d, a) are (40%, 10%), (40%, 20%), (40%, 30%), (30%, 20%) and (50%, 20%). The highest allowed answer for information valuation is 40% for all 5 scenarios.

Figure 6b shows the result of the Mutually Exclusive treatment. Average information valuation increases slightly with a , but the slope is statistically indistinguishable from zero ($p = 0.478$). At the individual level, 43.2% of participants have $\beta_a^i > 0.05$, which is not significantly more than the 29.7% with $\beta_a^i < -0.05$ (Pearson's χ^2 test on non-near-zero subjects, $p = 0.17$). These results provide little evidence that information valuations correctly respond to the alternative option, even when the computational requirements are minimal.

Interestingly, even though the correct information value does not depend on the default lottery, many participants' information valuations do. At the individual level, 51.4% of participants have $\beta_d^i < -0.05$, decreasing their valuations as d increases — significantly more than the 33.8% with $\beta_d^i > 0.05$ (two-proportion test, $p = 0.031$). Only 14.9% of participants correctly fall in the near-zero band ($|\beta_d^i| \leq 0.05$). The average information valuation is also more sensitive to d than to a , although the difference is not statistically significant ($p = 0.764$). These findings suggest that many people devalue information as the default lottery becomes more likely to win, even when such a response is not justified by the context. They are also consistent with the interpretation from D-Info (comparison) that the seemingly rational response of information valuation to d partly

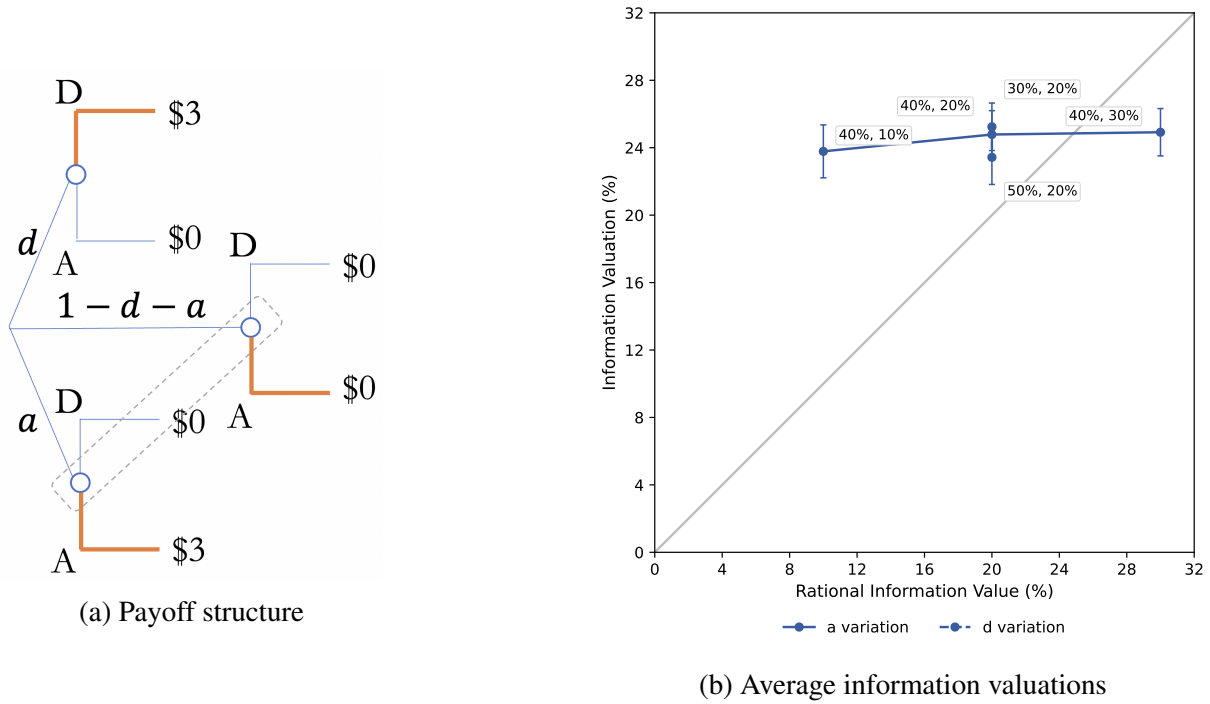


Figure 6: The D-Info (mutually exclusive) treatment: payoff structure and information valuations

Notes: Panel (a) depicts the payoff structure induced by the information in the D-Info (mutually exclusive) treatment. The straight branches represent uncertainty resolution, with the two nodes in the dashed rectangle being in the same information set; the letter on a straight branch represents its probability; the elbow branches are choice options; the optimal choices, shown in orange, induce a lottery. Panel (b) shows the average information valuations in the treatment. Each dot represents a scenario, with the tag next to it showing the d of that scenario. Error bars represent 95% confidence intervals.

reflects a satisficing heuristic.²¹

3.3.2 Multiple choices

To evaluate information, people need to integrate payoffs from multiple potential choices. This is true in my experiment because participants need to integrate the payoffs of choosing D and choosing A. It is also a general feature of information valuation because information adds value only if it changes people's choice with a positive probability. Prior research on contingent reasoning (Esponda and Vespa, 2014, 2021; Martínez-Marquina, Niederle, and Vespa, 2019) suggests

²¹In Appendix D.2, I present results from another treatment, Inconclusive Info, where information is not fully revealing, and rational information value does not depend on d . Similar to D-Info (mutually exclusive), average information valuation in that treatment incorrectly decreases in d . This result indicates that the heuristic identified about sensitivity to d extends beyond fully revealing information.

that integrating multiple potential payoffs can be challenging. This section examines whether this integration process becomes more difficult when the payoffs arise from multiple choices.

To test this hypothesis, I design a treatment where participants evaluate an insurance-like option that induces the same compound lottery as information but does not involve multiple choices in its valuation. Specifically, in the D-Insured treatment ($N = 142$), participants do not learn about a lottery's outcome in advance, nor are they asked to evaluate any information. Instead, they may receive an insurance option that allows them to use A as a back-up for D if they choose the latter. If they receive the insurance option and choose D, participants can win \$3 even if D fails, so long as A wins. (The option does not have any effect if A is chosen.) The two lotteries are independent and the (d, a) pairs covered are the same as in the 6 baseline scenarios in D-Info (comparison). Insurance valuations are elicited using the comparison-based method, where participants compare this option with an x percentage point increase in both lotteries' winning chances.

Having the insurance option leads to the compound lottery depicted in Figure 7a, which is the same as learning about D's outcome in the main experiment. Thus, the insurance value is the same as the information value, $(1 - d)a$. Unlike the treatments with information, however, participants in the D-Insured treatment should always choose D whether they have the insurance or not, so insurance valuation only requires integrating different payoffs from this single choice. Thus, comparing information valuation in the main experiment to insurance valuation in the D-Insured treatment allows me to test whether integrating payoffs from multiple choices is more difficult than integrating multiple payoffs from a single choice.

Figure 7b shows how average insurance valuation changes with d and a . Consistent with the rational comparative static in a , average insurance valuation increases with a ($p < 0.001$). This is in stark contrast with the average information valuation in the D-Info treatments, which is insensitive to a . The sensitivity to a in the D-Insured treatment is also reflected at the individual level: 50.0% of participants have $\beta_a^i > 0.05$, the largest share among all treatments presented so far. Moreover, 39.5% of participants refer to Lottery A in their incentivized advice, and essentially none of them mention the wrong comparative statics.

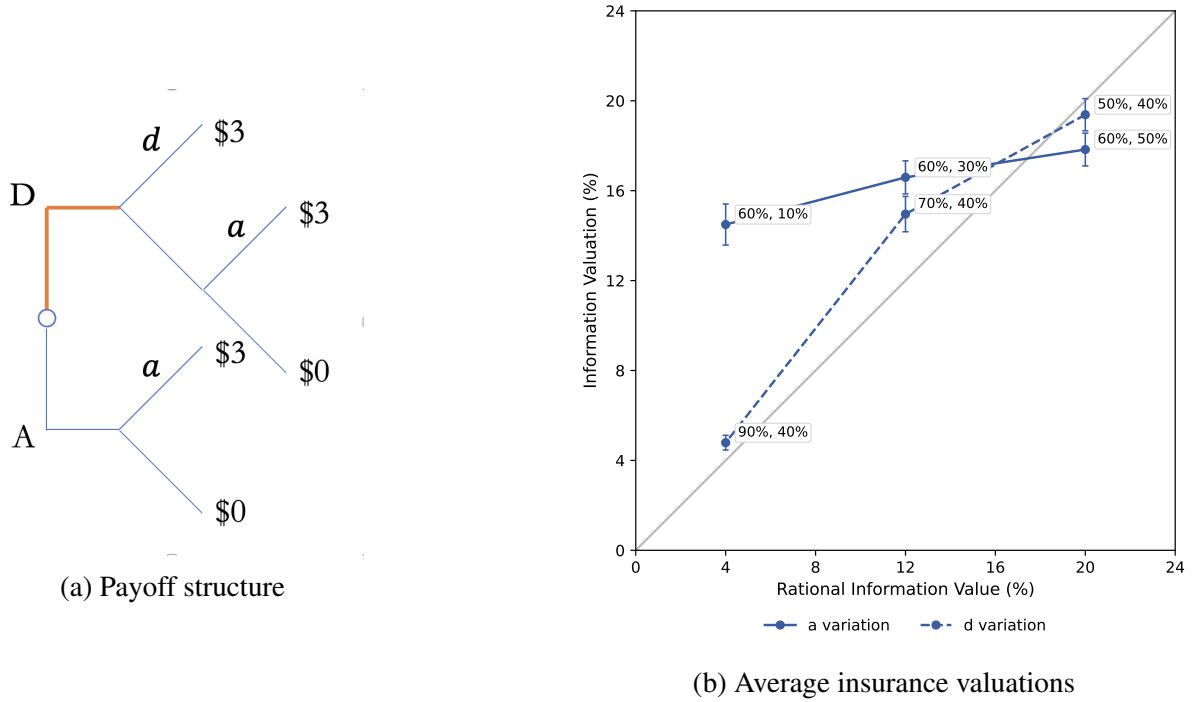


Figure 7: The D-Insured treatment: payoff structure and insurance valuations

Notes: Panel (a) depicts the payoff structure induced by the insurance in the D-Insured treatment. The straight branches represent uncertainty resolution; the letter on a straight branch represents its probability; the elbow branches are choice options; the orange branch is the optimal choice, which induces a compound lottery. Panel (b) shows the average insurance valuations in the treatment. Each dot represents a scenario, with the tag next to it showing the d and a of that scenario. Error bars represent 95% confidence intervals.

The different results from the information valuation experiments and the D-Insured treatment are consistent with the hypothesis that integrating payoffs from multiple choices is more challenging than integrating multiple payoffs from a single choice. However, the difference could also result from the back-up framing. To assess this possibility, I conduct a treatment, named A-Insured ($N = 156$), which keeps the back-up framing but asks participants to evaluate the option to insure Lottery A with D. Specifically, with the insurance option, participants who choose A can win \$3 even if it fails, so long as D wins. (The insurance option does not affect anything if D is chosen.) This insurance induces the same compound lottery as the information in the A-Info treatment (see Figure 8a), so their values are identical. However, the insurance changes the optimal decision: participants should choose A if it is insured and D if it is not. As a result, they still need to integrate payoffs from more than one choice to evaluate the insurance, which distinguishes it from the

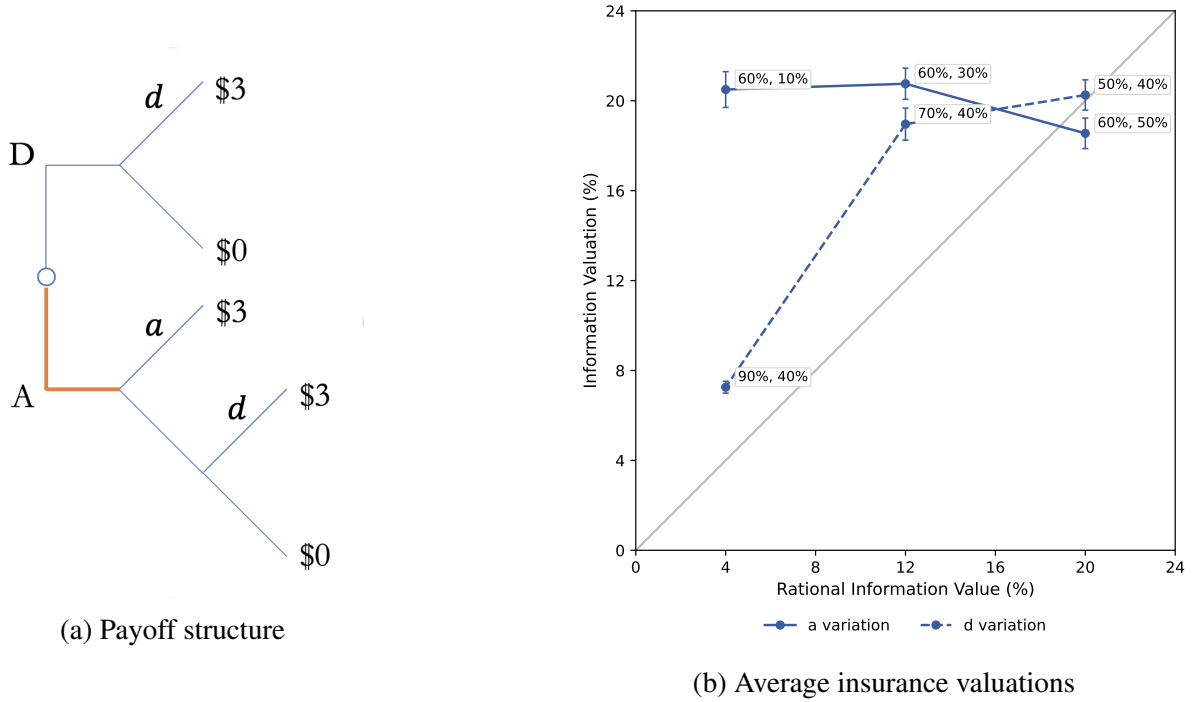


Figure 8: The A-Insured treatment: payoff structure and insurance valuations

Notes: Panel (a) depicts the payoff structure induced by the insurance in the A-Insured treatment. The straight branches represent uncertainty resolution; the letter on a straight branch represents its probability; the elbow branches are choice options; the orange branch is the optimal choice, which induces a compound lottery. Panel (b) shows the average insurance valuations in the treatment. Each dot represents a scenario, with the tag next to it showing the d and a of that scenario. Error bars represent 95% confidence intervals.

D-Insured treatment.

If integrating payoffs from multiple choices is what makes information valuation challenging, then we should not expect participants to perform better in the A-Insured treatment than in the A-Info treatment. If, on the other hand, the back-up framing is what helps participants in the D-Insured treatment, it should also help those in the A-Insured treatment.

Figure 8b shows average insurance valuations in the A-Insured treatment. The average valuation is decreasing in a ($p = 0.026$), showing no improvement from information valuations in the A-Info treatment. Similarly, the share of participants with $\beta_a^i < -0.05$ rises from 36.6% in the A-Info treatment to 44.9% in the A-Insured treatment, although the difference is not statistically significant (two-proportion test, $p = 0.10$). Among participants who give a directional claim in their incentivized advice, the wrong-direction shares dominate for d (22.9% vs. 5.9%) and are

roughly balanced for a (10.5% vs. 9.8%). Taken together, participants perform no better, and if anything worse, at evaluating insurance in the A-Insured treatment than at evaluating information in the A-Info treatment. This result suggests that the D-Insured improvement is not merely due to the back-up framing. It also supports the interpretation that having to integrate payoffs from multiple choices makes information valuation difficult.

3.4 Addressing multiple difficulties at once

The diagnostic treatments presented so far each use a single decision aid to address one component of the three-step difficulty. Even the most effective treatment, D-Insured, still leaves substantial deviations from the rational benchmark. This motivates a final set of “best-shot” treatments that combine several aids at once.

In the D-Info (best-shot) treatment ($N = 25$), participants choose between two mutually exclusive lotteries D and A (that is, at most one of the two lotteries can win) and may find out D’s outcome before the choice. As explained in Section 3.3.1, the rational information value in this setting is a , and its calculation does not require reductions of compound probabilities.

I elicit information valuation in 10 scenarios that vary the (d, a) pair: $(20, 10)$, $(30, 10)$, $(30, 20)$, $(40, 10)$, $(40, 20)$, $(40, 30)$, $(50, 10)$, $(50, 20)$, $(50, 30)$, $(50, 40)$. Valuations are elicited using the same direct method as in D-Info (direct): participants choose from 0%, 1%, ..., 39%, or “40% or more” to complete the sentence

Under this scenario, if I find out [Lottery D]’s outcome before choosing, my chance of choosing a [winning lottery] goes up by ____.²²

As discussed in Section 2.3.2, this method avoids the complexity of describing a comparison object.

Before evaluating the information, participants report a contingent plan specifying which lottery they would choose if the information is not available, and if the information is available and

²²“Lottery D” is referred to as “Investment A,” and “winning lottery” as “successful investment.”

reveals good or bad news about D. These contingent lottery choices are then shown on the screen when participants evaluate the information. Once the plan is made, the only cognitive step left for evaluating information is integrating payoffs from the planned choice of each contingency.

In this treatment, only one participant makes one lottery choice mistake. The near-perfect lottery choices mirror the result in the D-Info (strategy method) treatment and suggest that formulating optimal plans is not difficult once the contingencies are made explicit.

Despite the ease of formulating an optimal plan, participants who correctly evaluate information remain a small minority. Only 22.0% of answers across the 10 scenarios equal the rational value a .²³ This share is lower than the share of answers fitting the gap heuristic, previously also identified in the main experiment, that equates information value with $d - a$ (37.6%).²⁴ The prevalence of the gap heuristic contributes to wrong-signed sensitivity to a and spurious sensitivity to d in the aggregate, as is shown in Figure 9a and the regression analysis in Table A2.

I also conduct an A-Info version of this best-shot treatment ($N = 22$), where the evaluated information reveals the outcome of A, not D. All other design aspects and the rational benchmark remain the same. Since the information target is A, this treatment potentially draws more attention to a and makes its effect on information value more direct. This could improve information valuation.

Confirming this hypothesis, every participant states the optimal lottery choice plan, and the share of rational answers in A-Info (best-shot) is 44.1%, higher than the counterpart in D-Info (best-shot) by 22.1 percentage points ($p = 0.026$). This is consistent with the finding from the original A-Info treatment that making a more salient, or its effect on information value more direct, can improve information valuation when contingent thinking is easy. However, rational answers are still only modestly more prevalent than the gap heuristic (38.6%).²⁵ Average valuation in Figure 9b shows modestly positive sensitivity to a , but the regression coefficient on a reported in Table A2 is statistically indistinguishable from zero (0.21, s.e. 0.18). Moreover, the coefficient on d has

²³In the scenario where the rational benchmark is 40%, choosing “40% or more” is coded as exactly correct.

²⁴If I exclude the two scenarios where the rational answer and the gap heuristic answer coincide ($(d, a) \in \{(40, 20), (20, 10)\}$), these numbers become 17.0% and 36.5%.

²⁵The two numbers become 38.6% and 31.8% when excluding the two coinciding scenarios.

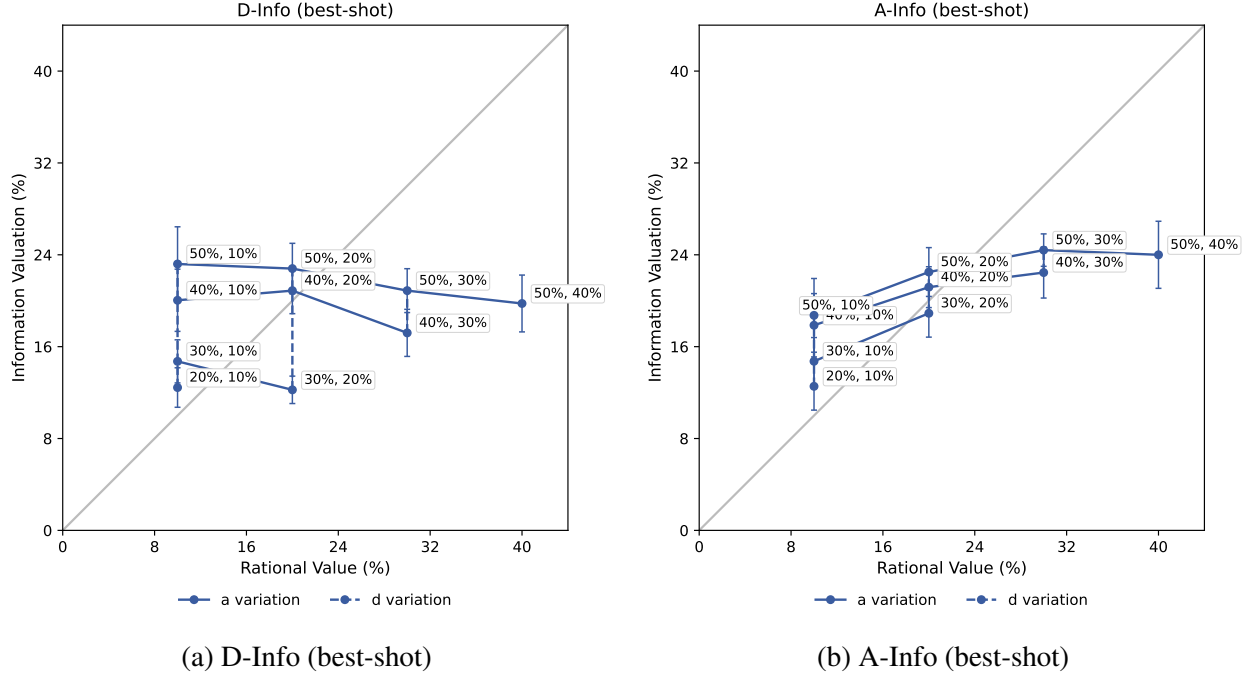


Figure 9: Average information valuations in the best-shot information treatments

Notes: Each panel plots the average information valuation against the rational value, which is a in both treatments because the two lotteries are mutually exclusive. Each dot represents one (d, a) scenario, with the label showing the values of d and a . Solid lines connect scenarios with the same d (varying a); dashed lines connect scenarios with the same a (varying d). The 45-degree gray line is the rational benchmark. Error bars represent ± 1 standard error.

the wrong sign (0.21, s.e. 0.10).

Taken together, the results indicate that even when the optimal plan of contingent lottery choices is made explicit and computational complexity reduced, information valuation remains challenging for a majority of people.

In contrast, valuations are correct in a majority of cases in a treatment that combines mutually exclusive lotteries and strategy-method choice elicitation with insurance valuation. In this treatment, named D-Insured (best-shot) ($N = 45$), participants evaluate an insurance option that allows them to use Lottery A as a back-up if they choose D and D fails, as in the original D-Insured treatment. The two lotteries are mutually exclusive, so the correct insurance value is simply a . Moreover, participants state the lottery they will choose with and without the insurance option before they evaluate it, making the already simple contingent choice explicit.²⁶

²⁶The statement that directly elicits insurance valuation is the following: Under this scenario, if I use [Lottery A] to

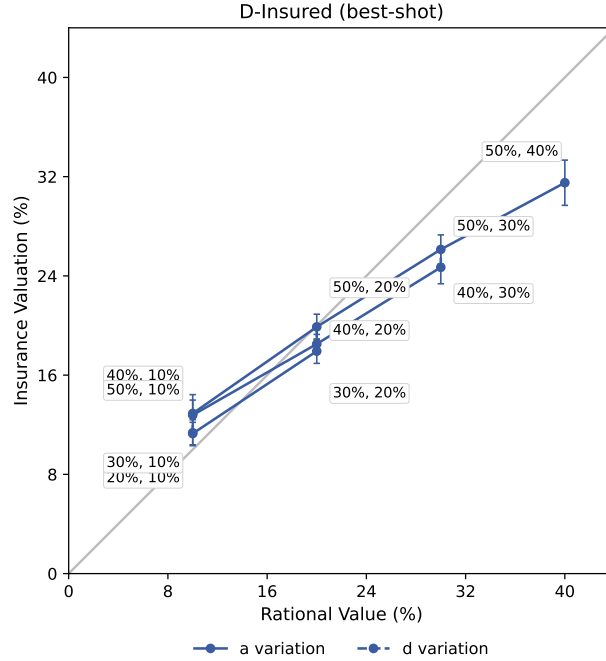


Figure 10: Average insurance valuations in the D-Insured (best-shot) treatment

Notes: This figure plots the average insurance valuation against the rational value, which is a because the two lotteries are mutually exclusive. Each dot represents one (d, a) scenario, with the label showing the values of d and a . Solid lines connect scenarios with the same d (varying a); dashed lines connect scenarios with the same a (varying d). The 45-degree gray line is the rational benchmark. Error bars represent ± 1 standard error.

In this treatment, all 45 participants correctly choose D both with and without insurance, and 66.0% of insurance valuations are exactly correct.²⁷ At the participant level, more than half of participants (24 out of 45) give correct valuation answers in all 10 scenarios, far above the corresponding shares in the best-shot D-Info (1 out of 25) and A-Info (5 out of 22) treatments. Average valuation shown in Figure 10 is therefore strongly sensitive to a , with a regression coefficient of 0.62 (s.e. 0.09). At the same time, the coefficient on d is small and statistically indistinguishable from zero (0.08, s.e. 0.04).

Results from the three best-shot treatments paint a mixed picture of the possibility of rational valuation. On the one hand, the strong result from the D-Insured (best-shot) treatment shows that, when each of the three conceptual steps is aided and valuation does not require integrating payoffs

back up [Lottery D], my chance of getting the bonus goes up by ____.

²⁷In contrast, only 24.4% of answers fit the gap heuristic $d - a$ (12.2% excluding the two scenarios where the heuristic coincides with the rational answer).

from multiple choices, a majority of participants can correctly value the induced payoff structure. On the other hand, the much weaker results from the best-shot information treatments suggest that integrating payoffs from multiple choices remains difficult even when contingent choices are made explicit and computational complexity is reduced. This limitation is important because integrating payoffs from multiple actions is inherent in information valuation.

4 Conclusion

4.1 Summary

This paper studies how demand for instrumental information responds to the decision problem it informs. In the main experiment, participants choose between two independent lotteries: a default lottery D with winning chance d and an alternative lottery A with winning chance $a < d$. They evaluate information that reveals D 's outcome before the choice. The rational value of this information is $(1 - d)a$: the probability that information changes the optimal choice, $1 - d$, multiplied by the gain in winning probability when it does, a . This rational benchmark is deliberately simple: it does not depend on participants' risk preferences or belief-updating rules.

Despite the simplicity of the environment, information valuations elicited using three different methods all deviate sharply from the rational benchmark. In the elicitation where participants compare the information with an x -percentage-point increase in both lotteries' winning chances, information valuation, measured by the x that makes a participant indifferent, decreases in d but remains insensitive to a on average. The insensitivity to a masks substantial individual-level heterogeneity, with some participants responding in the correct (positive) direction and others in the wrong direction. The sensitivity to d , though seemingly rational, partly reflects a heuristic that satisfices on the post-increase winning chance of D , as well as the low answer cap in the high- d scenario. In the elicitation where participants directly state how much the information increases their chance of choosing a winning lottery, the satisficing heuristic is muted because there is no explicit mention of an percentage increase in d and a . Instead, many substitute the winning-chance

gap $d - a$ for the value of information, producing wrong-signed average responses to both parameters. When participants state their willingness to pay for the information (in percentage-point terms), valuations are compressed across scenarios, consistent with loss aversion, which dampens sensitivity to both d and a . Taken together, the results suggest that participants who do not understand information value often resort to error-prone heuristics that depend on the elicitation context. At the same time, across the three elicitation methods, the behavior of participants who respond to a in the rational direction is fairly stable.

To investigate why information valuation is difficult for many participants, I conduct a series of diagnostic treatments. Results from these treatments show that making the alternative lottery's role more salient and direct, prompting participants to formulate contingent lottery-choice plans in advance, and simplifying the computation required for valuation each improve information valuation only modestly at best. Even when these aids are combined, only a minority of participants evaluate information correctly. The largest improvement occurs when, instead of information, participants evaluate an insurance-like option that induces the same payoff structure but removes the need to integrate payoffs from multiple possible choices. When combined with contingent-planning aids and simplified computation, the majority of participants evaluate the insurance option exactly correctly. These results imply that the difficulty of information valuation comes from multiple sources, so decision aids that address only one aspect have limited effects. They also point to payoff integration across multiple choices as a central difficulty, which is important because such integration is inherent in information valuation.

4.2 Discussion

The simplicity of the experimental design has both advantages and limitations. By using binary lotteries, fully revealing information, and elicitation mechanisms that measure information value in probabilistic terms, the experiment is able to separate cognitive difficulties specific to information valuation from preferences and biases in information use. At the same time, these design features inevitably make the environment abstract and remove domain knowledge and experience that may

help people in familiar contexts. Future research should examine whether the cognitive difficulties identified in this experiment also arise in richer field settings such as consumer search, medical testing, financial advice, hiring, or admissions.

The elicitation-dependence of the deviations also suggests that information demand is affected not only by the incentives for information acquisition, but also by how those incentives are framed. This finding suggests that research on information demand should pay close attention to contextual factors, especially the heuristics they may induce.

Results from the diagnostic treatments suggest that to achieve large improvements in information valuation, effective decision aids need to jointly address multiple sources of difficulty. In particular, integrating payoffs from multiple possible choices appears to be a key bottleneck. In the experiment, I remove this difficulty by replacing information with insurance, but such a transformation is unlikely to be available in many real decisions. Future research can explore other decision aids that address this difficulty.

The cognitive difficulties identified in this paper may also apply beyond information valuation. Many economic objects, ranging from incentive contracts and commitment devices to taxes and subsidies, derive value from changing future behavior, and evaluating them requires people to forecast possible actions and integrate payoffs across those actions. The difficulty of thinking through future choices and integrating their consequences may therefore generate biases in the valuation of, and support for, these interventions (see, e.g., Dal Bó et al., 2018; Carrera et al., 2022; Dreyfuss, 2025, for related evidence).

Biases in information valuation matter for individual welfare, as under-valuation may lead to neglect of risks and opportunities and over-valuation may lead to wasted resources on the acquisition of low-value information. Biases in the sensitivity of information valuation, the focus of this paper, may lead to context-dependent over- and under-valuation. Moreover, incorrect sensitivities may have important implications in market settings. For example, suppose the two lotteries in the experiment represent two competing products. If consumers can acquire information before purchase, the market share of the alternative product increases with the share of consumers who

acquire information.²⁸ If information demand is insensitive to or even decreases with the alternative product's prior value, then the marginal return to improving that product may be dampened. Of course, to validate such implications, further research on specific market settings is required.

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²⁸For example, if consumers can acquire information about the default product, the alternative's market share is $Pr(\text{Acquire information}) \cdot (1 - d)$. If consumers can acquire information about the alternative product, its market share is $Pr(\text{Acquire information}) \cdot a$.

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Appendix

A Additional tables

Table A1: Summary statistics of socio-demographics by treatment

	Comparison	Direct	WTP	A-Info	Strategy	Exclusive	D-Insured	A-Insured	D-Info (bs)	A-Info (bs)	D-Insured (bs)
Age	39.6	42.0	41.3	41.3	43.0	30.8	40.5	41.3	38.8	36.8	42.9
1 if female	0.54	0.51	0.44	0.46	0.55	0.49	0.52	0.44	0.32	0.59	0.44
1 if employed	0.72	0.73	0.71	0.72	0.64	0.57	0.68	0.73	0.84	0.77	0.78
1 if college degree	0.59	0.62	0.54	0.57	0.59	0.51	0.51	0.53	0.56	0.41	0.62
1 if income > 75k	0.34	0.51	0.20	0.43	0.40	0.11	0.39	0.32	0.64	0.36	0.58
1 if investing in stock	0.71	0.71	0.60	0.72	0.56	0.32	0.63	0.67	0.72	0.55	0.82
Info seeking (1-5)	3.86	4.01	—	3.87	4.03	4.18	3.89	3.96	3.88	4.27	4.02
Risk seeking (1-5)	3.07	3.25	3.07	3.12	2.99	3.45	3.04	3.21	3.52	2.91	3.04
Planning (1-5)	3.73	3.88	3.64	3.84	3.75	4.04	3.65	3.79	3.72	3.95	3.80
Subjects (<i>N</i>)	225	104	70	224	73	74	142	156	25	22	45

Notes: This table summarizes the socio-demographic questionnaire across all treatments. “Info seeking,” “risk seeking,” and “planning” are self-reported tendencies on five-point Likert scales (higher = stronger). The “info seeking” question is missing from the WTP treatment. “bs” = best-shot treatment.

Table A2: Within-treatment sensitivity to d and a , all treatments

Treatment	Rational benchmark	β_d (SE)	β_a (SE)	Subjects	Obs.	Within R^2
D-Info (comparison)	$(1 - d) a$	-0.349*** (0.015)	-0.043*** (0.015)	225	1,350	0.376
D-Info (direct)	$(1 - d) a$	+0.340*** (0.051)	-0.286*** (0.047)	104	1,352	0.148
D-Info (WTP)	$(1 - d) a$	-0.073*** (0.023)	+0.036** (0.017)	70	420	0.059
A-Info	$(1 - d) a$	-0.362*** (0.013)	-0.001 (0.018)	224	1,344	0.385
D-Info (strategy method)	$(1 - d) a$	-0.363*** (0.019)	+0.034 (0.028)	73	438	0.413
D-Info (mutually exclusive)	a	-0.091 (0.094)	+0.057 (0.070)	74	370	0.008
D-Insured	$(1 - d) a$	-0.371*** (0.017)	+0.077*** (0.023)	142	852	0.352
A-Insured	$(1 - d) a$	-0.335*** (0.016)	-0.058*** (0.021)	156	936	0.350
D-Info (best-shot)	a	+0.416*** (0.099)	-0.136 (0.149)	25	250	0.176
A-Info (best-shot)	a	+0.205* (0.101)	+0.207 (0.175)	22	220	0.145
D-Insured (best-shot)	a	+0.077* (0.040)	+0.617*** (0.089)	45	450	0.538

Notes: Each row reports the coefficients from the within-treatment regression $V_{is} = \alpha_i + \beta_d \cdot d_s + \beta_a \cdot a_s + \varepsilon_{is}$, where V_{is} is participant i 's information valuation in scenario s , d_s and a_s are the winning probabilities of D and A in scenario s , and α_i is a participant fixed effect. Simple scenarios ($d = 100$ or $a = 0$) are excluded from all regressions. Standard errors are clustered at the participant level. The rational benchmark column shows the theoretically correct information value, where $V = (1 - d) a$ for the standard treatments and $V = a$ for treatments where Lottery D and Lottery A are mutually exclusive. Under rationality, both treatments with benchmark $(1 - d) a$ predict $\beta_d \approx -\bar{a}/100$ and $\beta_a \approx (100 - \bar{d})/100$, while treatments with benchmark a predict $\beta_d = 0$ and $\beta_a = 1$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Distribution of individual-level sensitivity coefficients by treatment

	Comparison	Direct	WTP	A-Info	Strategy	Exclusive	D-Insured	A-Insured	D-Info (bs)	A-Info (bs)	D-Insured (bs)
<i>Sensitivity to d (β_d^i)</i>											
$\beta_d^i > 0.05$	4.0	64.4	11.4	0.9	1.4	33.8	2.1	3.2	60.0	40.9	22.2
$ \beta_d^i \leq 0.05$	9.3	7.7	34.3	4.5	0.0	14.9	7.0	7.1	12.0	22.7	66.7
$\beta_d^i < -0.05$	86.7	27.9	54.3	94.6	98.6	51.4	90.8	89.7	28.0	36.4	11.1
<i>Sensitivity to a (β_a^i)</i>											
$\beta_a^i > 0.05$	28.4	23.1	37.1	37.5	38.4	43.2	50.0	28.2	44.0	59.1	82.2
$ \beta_a^i \leq 0.05$	29.3	25.0	44.3	25.9	26.0	27.0	17.6	26.9	8.0	4.5	2.2
$\beta_a^i < -0.05$	42.2	51.9	18.6	36.6	35.6	29.7	32.4	44.9	48.0	36.4	15.6
Subjects (N)	225	104	70	224	73	74	142	156	25	22	45

Notes: For each participant i , I run an individual within-subject regression of information valuations on the two lottery winning chances, $V_{is} = \alpha_i + \beta_d^i \cdot d_s + \beta_a^i \cdot a_s + \varepsilon_{is}$, excluding simple scenarios ($d = 100$ or $a = 0$). The table reports the share of participants in each treatment (in %) whose estimated coefficient falls into each sign category, using a near-zero tolerance of $|\beta| \leq 0.05$. “bs” = best-shot treatment.

Table A4: Joint distribution of individual sensitivities (β_d^i, β_a^i) with Spearman correlation

Cell	Comparison ($N = 225$)	Comparison excluding $d = 90$ ($N = 225$)	Direct ($N = 104$)	WTP ($N = 70$)
$\beta_d^i > 0.05, \beta_a^i < -0.05$	3.6	14.7	45.2	4.3
$\beta_d^i > 0.05, \beta_a^i \leq 0.05$	0.4	2.7	14.4	1.4
$\beta_d^i > 0.05, \beta_a^i > 0.05$	0.0	5.8	4.8	5.7
$ \beta_d^i \leq 0.05, \beta_a^i < -0.05$	4.0	6.7	1.0	7.1
$ \beta_d^i \leq 0.05, \beta_a^i \leq 0.05$	3.6	14.2	5.8	18.6
$ \beta_d^i \leq 0.05, \beta_a^i > 0.05$	1.8	3.6	1.0	8.6
$\beta_d^i < -0.05, \beta_a^i < -0.05$	32.9	18.7	5.8	7.1
$\beta_d^i < -0.05, \beta_a^i \leq 0.05$	24.0	14.2	4.8	24.3
$\beta_d^i < -0.05, \beta_a^i > 0.05 \checkmark$	29.8	19.6	17.3	22.9
Spearman ρ	+0.215***	+0.202***	+0.810***	+0.066
(p -value)	(0.001)	(0.002)	(0.000)	(0.590)

Notes: Each cell in the top block reports the percentage of participants in the indicated D-Info treatment whose individual sensitivities (β_d^i, β_a^i) , estimated from equation (3), fall in the indicated joint bin. The three β_d^i -bins partition each treatment exactly, as do the three β_a^i -bins, using a near-zero tolerance of $|\beta| \leq 0.05$. The “Comparison excluding $d = 90$ ” column re-estimates the coefficients for Comparison treatment after also excluding the $d = 90$ scenario, where the answer-range cap is 10%. The rational (“both correct”) joint bin $(\beta_d^i < -0.05, \beta_a^i > 0.05)$ is marked with $\checkmark$. The Spearman correlation ρ is computed between $-\beta_d^i$ and β_a^i , so that a *positive* ρ means that participants’ correctness on d and on a move together: a participant who is correct on one parameter tends to be correct on the other, and a participant who is wrong on one tends to be wrong on the other (large mass in the top-right and bottom-left of the joint table). A negative ρ would indicate the opposite: correctness on one parameter pairs with wrongness on the other. The two-sided p -value is from the t -approximation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Comparative statics of information valuation mentioned in advice

% of participants who write that info valuation should...	Comparison	Direct	WTP	A-Info	Strategy	Exclusive	D-Insured	A-Insured	D-Info (bs)	A-Info (bs)	D-Insured (bs)
<i>Comparative statics on d</i>											
decrease in <i>d</i>	20.3 ✓	27.9 ✓	39.1 ✓	20.0 ✓	21.9 ✓	17.9	21.6 ✓	5.9 ✓	44.0	31.8	57.8
mentions Lottery D without direction	10.8	9.6	5.8	13.2	11.0	9.0	10.8	9.2	20.0	4.5	4.4
increase in <i>d</i>	14.9	37.5	2.9	10.0	5.5	7.5	10.8	22.9	20.0	18.2	4.4
<i>Comparative statics on a</i>											
increase in <i>a</i> ✓	11.3	25.0	13.0	21.4	8.2	10.4	33.8	9.8	32.0	45.5	51.1
mentions Lottery A without direction	7.2	4.8	2.9	9.5	6.8	4.5	5.0	9.8	4.0	9.1	2.2
decrease in <i>a</i>	12.6	28.8	10.1	11.8	5.5	4.5	0.7	10.5	16.0	18.2	2.2
Subjects (<i>N</i>)	222	104	69	220	73	67	139	153	25	22	45

Notes: For each participant, the open-ended advice/reasoning text is coded into binary indicators using Claude Opus 4.6 with the codebook described in Appendix B. *d* and *a* refer to the winning chances of the default investment (Investment A) and the alternative investment (Investment B), respectively. The ✓ label indicates the correct comparative statics. “bs” = best-shot.

Table A6: Cross-tabulation of behavioral type and advice direction on d

Advice category	$\beta_d^i < -0.05$	$ \beta_d^i \leq 0.05$	$\beta_d^i > 0.05$
Panel A: Comparison (total $N = 222$; $N_- = 193$, $N_0 = 21$, $N_+ = 8$)			
Mentions “decrease in d ”	43 (22%)	1 (5%)	1 (12%)
No directional claim on d	122 (63%)	16 (76%)	6 (75%)
Mentions “increase in d ”	28 (15%)	4 (19%)	1 (12%)
<i>Spearman correlation:</i> $\rho = +0.103$ ($p = 0.127$)			
Panel B: Comparison (excluding $d = 90$) (total $N = 222$; $N_- = 116$, $N_0 = 55$, $N_+ = 51$)			
Mentions “decrease in d ”	36 (31%)	6 (11%)	3 (6%)
No directional claim on d	67 (58%)	43 (78%)	34 (67%)
Mentions “increase in d ”	13 (11%)	6 (11%)	14 (27%)
<i>Spearman correlation:</i> $\rho = +0.284^{***}$ ($p = 0.000$)			
Panel C: Direct (total $N = 104$; $N_- = 29$, $N_0 = 8$, $N_+ = 67$)			
Mentions “decrease in d ”	16 (55%)	3 (38%)	10 (15%)
No directional claim on d	6 (21%)	5 (62%)	25 (37%)
Mentions “increase in d ”	7 (24%)	0 (0%)	32 (48%)
<i>Spearman correlation:</i> $\rho = +0.373^{***}$ ($p = 0.000$)			
Panel D: WTP (total $N = 69$; $N_- = 38$, $N_0 = 23$, $N_+ = 8$)			
Mentions “decrease in d ”	22 (58%)	4 (17%)	1 (12%)
No directional claim on d	16 (42%)	18 (78%)	6 (75%)
Mentions “increase in d ”	0 (0%)	1 (4%)	1 (12%)
<i>Spearman correlation:</i> $\rho = +0.444^{***}$ ($p = 0.000$)			

Notes: Each cell reports the number of participants in the indicated (β_d , advice on d) cell, with the column share in parentheses. The behavioral classification (columns) uses the same 3-way coding as in Appendix Table A3, with a near-zero band of $|\beta| \leq 0.05$. The advice classification (rows) collapses “mentions d , no direction” and “no mention of d ” into a single neutral middle row, leaving three ordered categories that mirror the behavioral coding. Spearman’s ρ is computed by coding both axes as $-1/0/+1$; the two-sided p -value is from the t -approximation. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A7: Cross-tabulation of behavioral type and advice direction on a

Advice category	$\beta_a^i < -0.05$	$ \beta_a^i \leq 0.05$	$\beta_a^i > 0.05$
Panel A: Comparison (total $N = 222$; $N_- = 89$, $N_0 = 63$, $N_+ = 70$)			
Mentions “decrease in a ”	20 (22%)	4 (6%)	4 (6%)
No directional claim on a	61 (69%)	58 (92%)	50 (71%)
Mentions “increase in a ”	8 (9%)	1 (2%)	16 (23%)
<i>Spearman correlation:</i> $\rho = +0.258^{***}$ ($p = 0.000$)			
Panel B: Direct (total $N = 104$; $N_- = 54$, $N_0 = 26$, $N_+ = 24$)			
Mentions “decrease in a ”	25 (46%)	4 (15%)	1 (4%)
No directional claim on a	21 (39%)	17 (65%)	10 (42%)
Mentions “increase in a ”	8 (15%)	5 (19%)	13 (54%)
<i>Spearman correlation:</i> $\rho = +0.442^{***}$ ($p = 0.000$)			
Panel C: WTP (total $N = 69$; $N_- = 12$, $N_0 = 31$, $N_+ = 26$)			
Mentions “decrease in a ”	3 (25%)	3 (10%)	1 (4%)
No directional claim on a	8 (67%)	26 (84%)	19 (73%)
Mentions “increase in a ”	1 (8%)	2 (6%)	6 (23%)
<i>Spearman correlation:</i> $\rho = +0.281^{**}$ ($p = 0.019$)			

Notes: Each cell reports the number of participants in the indicated (β_a , advice on a) cell, with the column share in parentheses. The behavioral classification (columns) uses the same 3-way coding as in Appendix Table A3, with a near-zero band of $|\beta| \leq 0.05$. The advice classification (rows) collapses “mentions a , no direction” and “no mention of a ” into a single neutral middle row, leaving three ordered categories that mirror the behavioral coding. Spearman’s ρ is computed by coding both axes as $-1/0/+1$; the two-sided p -value is from the t -approximation. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

B LLM-based coding of open-ended responses

This appendix describes the procedure used to classify the open-ended advice in D-Info (comparison), D-Info (WTP) and reasoning in D-Info (direct) responses into themes for Table 2 in the main text. The procedure uses a large language model to inductively discover, consolidate, and apply a coding scheme. All calls are made to Anthropic’s Claude Opus 4.6 model, with fixed seeds and deterministic settings to ensure reproducibility.

B.1 Three-phase pipeline

Phase 1: Discovery. For each treatment separately, I draw three independent random samples of 50 responses (with three different fixed random seeds). For each sample, I prompt the model to identify between 5 and 10 distinct “reasoning patterns or decision rules” that participants articulate, where each theme must be a substantive decision rule rather than a topic, and each must be supported by at least two verbatim example quotes drawn from the sample. For each theme, the model returns a name, a one-sentence definition that another coder could apply unambiguously, and two verbatim example quotes with subject identifiers. I run this discovery step three times per treatment and compare the proposed themes across runs: themes that appear in all three runs are considered robust; themes that appear in only one run are considered noise.

This discovery step uses a slight nonzero temperature (0.3) to allow the three runs to vary, which serves as a robustness check on the proposed themes. The model is not shown any aspect of the theoretical framework or hypotheses at any point in the discovery step.

Phase 2: Consolidation. The nine discovery outputs (three runs $\times$ three treatments) are combined into a single prompt that asks the model to merge near-duplicate themes across treatments and runs into a unified codebook of 8 to 12 themes. The consolidation prompt explicitly instructs the model to keep treatment-specific themes when they are substantively important (so that, for example, the loss-aversion theme that only appears in the WTP treatment is preserved) and to drop themes that appear in only one of the nine discovery outputs (which are likely to be artifacts). The

result is a 12-theme master codebook with names, one-sentence definitions, and a list of treatments in which each theme primarily appears.

I review the consolidated codebook by hand to verify that the definitions are clear and that no important distinctions have been merged away. The complete codebook is reproduced in Table A8 below.

Phase 3: Classification. The master codebook is then applied to every response in all three treatments. For each response, the model is prompted with the codebook and asked to return a JSON object indicating, for each theme, whether the theme is present (true) or absent (false) in the response. A theme is marked present only if the response clearly expresses the reasoning pattern in its definition; ambiguous cases are marked false. Classification is done one response at a time at temperature 0, so that the classification of one response cannot be influenced by other responses or by stochastic variation across calls.

Second-pass reclassification. After the initial classification step, two of the discovered themes admit further substantive sub-categorization. I run a separate reclassification pass for each, restricted to the subset of responses coded as expressing the parent theme. Each pass uses temperature 0.0 and asks the model to return a JSON object with mutually exclusive flags.

1. *Spread-direction split.* The theme `spread_between_default_and_alternative` can describe two opposite reasoning patterns: “information is more valuable when the spread is large” (the gap heuristic) and “information is more valuable when the spread is small” (closer to correct). Because the direction of this reasoning is theoretically important, I ask the model to classify each spread-flagged response into `spread_large_increases_info`, `spread_small_increases_info`, or `unclear`. The two sub-themes are reported in Table 2.
2. *Threshold-rule split.* The theme `probability_threshold_rule` captures responses that apply an explicit numeric cutoff on D’s chance, but without specifying *what* the cutoff

is doing. I run two complementary passes over the threshold-flagged responses. The first determines the direction of the cutoff: `threshold_low_d_high_info` (information valuable when D 's chance is below the cutoff, the rational direction) versus `threshold_high_d_high_info` (information valuable when D 's chance is above the cutoff). The second determines the underlying *type* of decision rule the cutoff expresses. Two of these capture forms of *satisficing on a winning chance*, differing in which chance the cutoff is on: `upgrade_satisficing` — satisficing on the post-increase winning chance — where the cutoff is on D 's post-upgrade chance, so the participant takes the upgrade only when it pushes D above the target; and `prior_winning_chance_satisficing` — satisficing on the prior winning chance — where the cutoff is on D 's prior chance and the participant prefers the report when D is uncertain or low (the rational direction). The third sub-theme, `wrong_direction_threshold`, captures cutoffs implying the report is more valuable when D is already high. The satisficing sub-themes are reported in Table 2.

B.2 The codebook

Table A8: Master codebook used for classification

Theme	Definition
report_eliminate_uncertainty	Participant values the information because it definitively reveals whether D succeeds or fails, enabling a fully informed choice rather than relying on probabilistic guessing.
switch_to_alternative_if_default_fails	Participant explicitly describes a contingent switching strategy: learn D's outcome, choose D if it succeeds, and switch to A as a fallback if D fails.
high_default_probability_favors_info	Participant reasons that information about D is more valuable when D already has a high probability of success.
high_probability_reduces_info_value	Participant reasons that when D's probability is already very high (or outcomes are near-certain), information is less valuable because the decision is already clear.
uncertainty_or_moderate_probability_increases_info_value	Participant reasons that information is most valuable when probabilities are moderate (near 50%) or otherwise uncertain.
spread_between_default_and_alternative	Participant bases the value of information on the difference (spread) between the success probabilities of D and A. The direction of this reasoning is split into two sub-themes (large spread vs small spread).
alternative_quality_as_fallback	Participant reasons that the value of knowing D's outcome depends on A's probability of success — information is more useful when A provides a viable alternative if D fails.
probability_threshold_rule	Participant applies an explicit numeric probability cutoff (e.g., 50%, 60%, 70%) on D's chance to decide whether to take the report or the alternative.
always_prefer_upgrade_or_reject_info	Participant adopts a blanket rule to always choose the percentage upgrade over the report, or concludes the report is never worth buying.
always_prefer_report	Participant adopts a blanket rule to always choose the report over the upgrade.
chip_cost_lottery_tradeoff	Participant explicitly recognizes that spending points on the report reduces their lottery winning probability, framing the decision as a direct tradeoff between information value and lottery equity.
gut_feeling_intuition	Participant reports relying on gut feeling, instinct, or intuition rather than a systematic decision rule.

B.3 Prevalence of each theme by treatment

The table below reports the share of participants in each treatment whose response is flagged for each of the 12 master themes and for the relevant second-pass sub-themes. The main-text table (Table 2) shows a curated subset; this full version is reproduced here for completeness.

Table A9: Prevalence of all 12 reasoning themes by treatment

% of participants whose reasoning expresses...	Comparison	Direct	WTP
Report eliminates uncertainty	14.0	14.4	4.3
Switch to A if D fails (correct contingent plan)	5.0	8.7	0.0
High $a \Rightarrow$ info more valuable (correct direction on a)	8.6	25.0	2.9
High $d \Rightarrow$ info more valuable	8.6	17.3	1.4
High prior $\Rightarrow$ info less valuable	14.0	4.8	40.6
Uncertainty/moderate prior $\Rightarrow$ info more valuable	10.8	18.3	26.1
Mentions spread between D and A	14.4	34.6	13.0
large spread $\Rightarrow$ info more valuable (gap heuristic)	4.5	21.2	1.4
small spread $\Rightarrow$ info more valuable	6.8	9.6	10.1
spread mentioned without direction	4.1	8.7	1.4
Probability threshold rule (any cutoff on d)	26.6	18.3	21.7
Satisficing on post-increase winning chance	13.1	0.0	0.0
Satisficing on prior winning chance	3.2	1.0	14.5
Wrong-direction cutoff (e.g., “info more useful when d is high”)	2.7	4.8	2.9
Cutoff with unclear interpretation	9.5	12.5	5.8
Always prefer upgrade / reject info	12.2	6.7	7.2
Always prefer report	8.6	2.9	1.4
Cost concern (loss aversion)	0.9	0.0	46.4
Gut feeling / intuition	18.0	2.9	8.7
Subjects (N)	222	104	69

Notes: Full version of Table 2 reporting prevalence for all 12 master themes from the codebook (see Appendix B) plus the second-pass sub-themes. Indented rows are sub-themes of the immediately preceding parent and partition the participants flagged for that parent (e.g., a participant flagged for `probability_threshold_rule` is also classified into one of the four indented threshold sub-themes). Themes are non-exclusive across the master themes: a single response can be coded for multiple master themes.

C LLM-based coding of advice direction

This appendix describes the procedure used to classify the directional content of participants' open-ended advice/reasoning responses for Table A5 and the cross-tabulations in Tables A6 and A7. The classification is performed by Claude Opus 4.6 with deterministic settings (temperature = 0) using five elicitation-specific codebooks. Each subject is classified using the codebook that matches the elicitation method and the type of target option they were asked to value.

C.1 Six indicators per response

For each open-ended response, the model returns six binary indicators describing whether and how the participant says the value of the target option depends on Investment A (Lottery D)'s and Investment B (Lottery A)'s winning chances:

- `mentions_a_lower_more_valuable`: target option more valuable when Investment A's chance is *lower*.
- `mentions_a`: participant invokes Investment A's chance as relevant but takes no directional stance.
- `mentions_a_higher_more_valuable`: target option more valuable when Investment A's chance is *higher*.
- `mentions_b_higher_more_valuable`: target option more valuable when Investment B's chance is *higher*.
- `mentions_b`: participant invokes Investment B's chance as relevant but takes no directional stance.
- `mentions_b_lower_more_valuable`: target option more valuable when Investment B's chance is *lower*.

The three indicators for each investment are mutually exclusive: at most one is set per investment. If a directional indicator is set, the non-directional `mentions_a` or `mentions_b` is not also set.

Mutual exclusivity is enforced as a post-processing step.

Throughout the codebooks, the model is told only about the participant-facing labels (“Investment A”, “Investment B”) and the target option being valued. It is not told which investment is the default, what the rational benchmark is, or what the experimenter expects.

C.2 Five elicitation-specific codebooks

The five codebooks share the six indicator names and the same rule structure, but each is tailored to the framing of its treatment:

1. *Information valuation under Comparison Elicitation* (D-Info comparison, A-Info, Strategy, Mutually Exclusive): target is the *report* revealing one of the investments’ outcomes; the alternative is an *upgrade* that increases both investments’ winning chances.
2. *Information valuation under WTP* (D-Info WTP): target is the report; the alternative is keeping the *chips* that would otherwise be spent on it.
3. *Insurance valuation under Comparison Elicitation* (D-Insured, A-Insured): target is the *back-up* option; the alternative is the upgrade.
4. *Information valuation under Direct Elicitation* (D-Info direct, D-Info best-shot, A-Info best-shot): target is the information; the participant states its value directly with no alternative to compare against.
5. *Insurance valuation under Direct Elicitation* (D-Insured best-shot): target is the back-up; direct statement, no alternative.

The two “comparison” codebooks explicitly instruct the model to invert the sign when the participant prefers the alternative under some condition: “if the upgrade is preferred when condition X holds, infer the target option is *less* valuable under X .” For the two insurance treatments, the prompt also specifies which investment serves as the back-up (Investment B in D-Insured and D-Insured best-shot; Investment A in A-Insured) so that statements like “the back-up’s chance is over 40%” are attributed to the correct investment.

C.3 Gap and direction rules

Two refinements help the model handle reasoning that involves the difference between the two investments' chances or that uses terse directional language:

- *Gap rule.* When a participant invokes the gap, difference, or spread between Investments A and B (e.g., “when A is much higher than B”, “when they are close”), the codebook treats this as mentioning both investments and decomposes any directional claim about the gap into the corresponding directions for A and B. For example, “information is more valuable when the difference is large” triggers `mentions_a_higher_more_valuable` and `mentions_b_lower_more_valuable` jointly.
- *Direction-marking rule.* A directional indicator is set whenever the participant uses directional language (“higher”, “lower”, “over $X\%$ ”, “around $Y\%$ ”, and so on) attached to an investment's chance and links it to the value of the target option. The non-directional `mentions_a` or `mentions_b` is reserved for cases where the participant invokes the chance without taking a clear directional stance.

D Additional treatments

D.1 Advisee treatment

In this section, I report on the design and results of the D-Info (advisee) treatment ($N = 51$). This treatment is identical to the D-Info (comparison) treatment with the main exception being that participants evaluate advice provided by previous D-Info participants after going through the instructions, but prior to encountering the scenarios. Specifically, each participant in the advisee treatment reviews two pieces of advice written by participants from earlier D-Info treatments where the rational benchmark is $(1 - d)a$. One piece of advice correctly argues that information valuation should increase in a , whereas the other incorrectly asserts a reverse relationship. Before they begin to evaluate information across the six scenarios, participants indicate which piece of advice they find more persuasive or if they deem them equally convincing. The participants in the advisee treatment do not encounter the simple scenario with $a = 0$ and are not asked to offer advice.

The number of participants identifying the correct advice as more convincing (19) is significantly larger (Pearson's χ^2 test, $p = 0.059$) than those misidentifying (9). This outcome implies that, on average, participants possess some capacity to discern the correct relationship between information valuation and the alternative lottery once the arguments have been laid out for them. However, this ability does not translate to any significant improvement in the sensitivity of average information valuation to a (see Figure A1). At the participant level, 31.4% of participants exhibit information valuations that increase monotonically with a . This percentage exceeds the 20.4% in the D-Info treatment. Nevertheless, this difference is solely driven by participants who rate the two pieces of advice as equally convincing. Among these participants, 43.5% display this correct comparative statics, whereas only 21% and 22.2% of those favoring the correct and incorrect advice, respectively, exhibit the same. Taken together, these results indicate that exposure to the advice of others has a limited effect on information valuations.

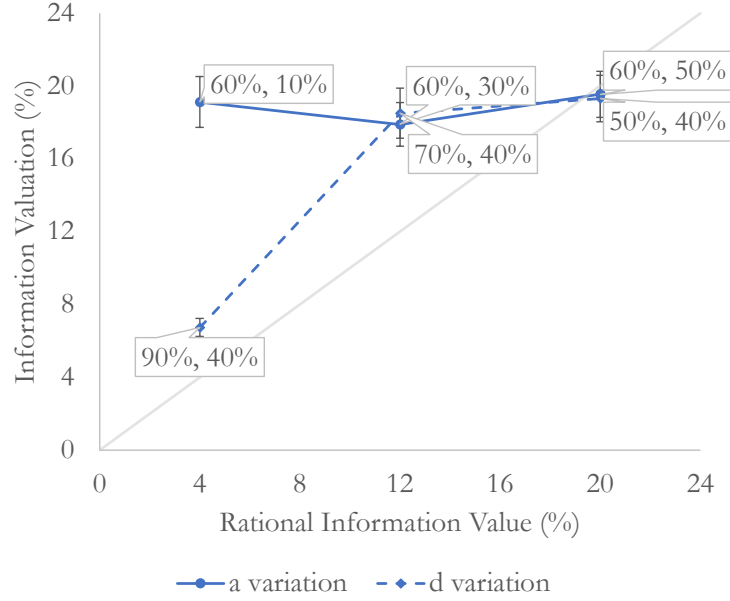


Figure A1: Information valuations in the D-Info (advisee) treatment

Notes: This figure shows the average information valuations in the D-Info (advisee) treatment. Each dot represents a scenario, with the tag next to it showing the d and a of that scenario. Error bars represent 95% confidence intervals.

D.2 Inconclusive Info treatment

In this section, I report on the Inconclusive Info treatment ($N = 86$). Similar to the D-Info (mutually exclusive) treatment, results of this treatment show that information valuations respond to the default lottery even when they shouldn't. This treatment also demonstrates that the bounded rationality in information valuation persists even when the calculation doesn't involve reducing compound probabilities.

In the Inconclusive Info treatment ($N = 86$), bad news about Lottery D's outcome is conclusive but good news is not—the information reports “Lottery D wins” with probability $d' > d$. To keep the number of parameters constant, I simplify the setting by making D and A perfect complements—A wins if and only if D loses. Same as in the D-Info treatment, participants should choose D if the information says it wins and choose A otherwise. This strategy leads to a lottery depicted in Figure A2a. The total winning chance of following this strategy is $d + 1 - d'$, which implies that the information value should be $1 - d'$. Note that this expression does not involve any reduction

